

RHG - Technical Standards



2. Technical Standards for Mechanical, Electrical & Plumbing

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LIST OF ABBREVIATIONS USED IN THESE STANDARDS:

A&D	RADISSON HOTEL GROUP BRAND'S ARCHITECTURAL AND DESIGN GUIDELINES
AED	AREA ENGINEERING DIRECTOR
AHU	AIR HANDLING UNIT
AV	AUDIO VISUAL SYSTEMS
AVP	RADISSON HOTEL AREA VICE PRESIDENT
BMS	BUILDING MANAGEMENT SYSTEM
BOH	BACK OF HOUSE
D&T	RADISSON HOTEL GROUP DESIGN & TECHNICAL SERVICES
DHW	DOMESTIC HOT WATER
F&B	RADISSON HOTEL FOOD AND BEVERAGE DEPARTMENT
FCU	FANCOIL UNIT
FF&E	FURNITURE, FIXTURES AND EQUIPMENT
FOH	FRONT OF HOUSE
H&S	HEALTH AND SAFETY
M&E	MEETINGS AND EVENTS
MEP	MECHANICAL, ELECTRICAL AND PLUMBING INSTALLATION
MUR	MOCK-UP ROOM
OSE	OPERATING SUPPLIES & EQUIPMENT
OWNERS	OWNERS/DEVELOPERS AND THEIR PROJECT TEAM INCLUDING THE OWNER'S REPRESENTATIVE/PROJECT MANAGER, ARCHITECT, INTERIOR DESIGNER, ENGINEERS, ETC.
PAP	PROJECT APPROVAL PROCESS
PUA	PUBLIC AREAS
PM	PROJECT MILESTONES
PMS	PROPERTY MANAGEMENT SYSTEM
POS	POINT OF SALES
PUA	PUBLIC AREAS
R&C	RADISSON HOTEL GROUP BRAND'S ROOMS AND CORRIDOR GUIDELINES
RHG	RADISSON HOTEL GROUP
SHOP	RADISSON HOTEL GROUP CORPORATE PURCHASING INFORMATION NETWORK
STANDARDS	TECHNICAL STANDARDS
TSA	TECHNICAL SERVICES AGREEMENT

01. INTRODUCTION

01.01 Purpose of This Document

This document sets out the Technical Brand Standards for Mechanical, Electrical and Plumbing (the “Standards”) required for all RHG brands and are provided to assist owners/developers and the project team (the “Owners”) to plan, specify and execute the proposed works. In addition, as the project progresses, a series of detailed supporting documentation will be made available for further consultation with the project team.

The Standards aim to provide the project team involved in the design, construction and renovation of RHG’s with sufficient information to ensure that the project is executed in compliance with the RHG brands values. Everyone involved in the construction and renovation process is expected to have read and understood these Standards. These Standards do not aim to cover every single detail of the design and construction of RHG properties, but rather set out the minimum requirements to be followed and are subject to the statutory regulations and requirements of local government, building control, planning authorities, etc.

01.02 Scope of This Document

The main objectives of providing and implementing the Standards can be summarized as follows:

- Ensure the safety and security of our guests and employees,
- Embody and maintain the corporate identity of the RHG brands,
- Deliver consistent levels of quality, comfort and service to our guests,
- Increase operational profits by efficient use of space, reduced consumption of utilities, selection of reliable, durable and well-designed products which are of an approved quality, meet the operator’s functional requirements and give an optimum balance of style and price,
- Establish specification requirements and acquaint the owners and developers with their construction and design obligations towards RHG,
- Increase awareness of sustainable development good practice in the design and construction of the hotel.

01.03 Copyright

These Standards, including the documents referred to, are a confidential and proprietary document supplied by RHG following the agreement between the Owner and RHG. Any alteration to the content of printed material and unauthorized copying of the contents is prohibited. This document may not be reproduced, disclosed, reproduced or distributed outside the project team and the contents may only be copied and merged into documents which are being created to construct or refurbish a RHG branded hotel by the Owners.

01.04 National/Local Regulations

All applicable National/Local statutory regulations and/or requirements must be fully complied with in the planning and execution of the project. In the event of the Standards exceeding the requirements of the applicable statutory regulations, the Standards will prevail, unless specifically prohibited by law. Irrespective RHG will have no liability or obligation in case of non-compliance with statutory regulations as this remains the Owner’s responsibility.

If the project is in an area not regulated by building codes, and/or the applicable codes do not adequately cover all aspects of the work then RHG must be informed which codes are proposed to govern the design of the works. However, the requirements of such proposed codes must as a minimum be in compliance with European Norms (EN). In addition, there may also be a requirement to comply with NPFA regulations, in relation to fire protection issues, within certain locations due to local market demands. All sprinkler installations are to be NFPA compliant.

01.05 Deviations

It is assumed that the Standards shall be complied with in their entirety. However, and only in particular circumstances should the adoption of any Standard be considered inappropriate, or unfeasible, then prior permission to deviate from the Standards must be sought from RHG by the Owner before any implementation of design or the works.

Only in the case of written request can alternatives be agreed to be adopted or for specific provisions to the Standards to be omitted for the specific project. This is a case per case assessment and under no circumstances will this be interpreted as

being automatically applicable to any other project. All agreed deviations will be recorded within a signed off deviations schedule that shall be used to track all agreed changes during the delivery of the project.

02. HVAC (Heating Ventilation & Air Conditioning)

02.01. General

The design of basic heating, ventilation, and air conditioning (HVAC) systems will vary widely due to the different climates in which the properties are located and a suitable highly efficient and sustainable air conditioning system that is best suited to the location is to be designed by specialists. Agreed system to be designed and installed by qualified party as per international codes, ASHRAE and RHG guidelines.

Hotels should be comfort cooled and heated by means of a 4-pipe ducted air/water system complete with chilled and hot water coils where applicable as per yearly weather temperature changes. However, it is recognized that some properties may not require cooling, some properties may only require winter heating and many others may require only cooling throughout the year. However, the proposed approach will be agreed with RHG in line with local market.

In certain locations and subject to RHG D&T approval a VRV (Variable Refrigerant Volume) system can be considered where the outdoor condensing units can be positioned to be visually and audibly isolated from the public. Type, configuration, location and control of indoor fan-coil VRV components in guest rooms shall be approved by RHG. The VRV system should have a leak detection alarm that is linked to the BMS and indicates the specific location of leak so that minimum no. of guestrooms needs to be evacuated. VRV system can be 2 or 3 Pipe system depending on the property and its location.

The room fan coil unit to be supplied with strainer, drain pan, fan with frequency variable speed motor and to be mounted in guestroom entrance lobby ceiling with easy access for maintenance and/or replacement. The position of FCU supplies air to avoid blowing directly on the bed. Coordinate supply grille and access panel with Interior Design.

Demand Controlled Air Conditioning with use of occupancy sensors to provide automatic setback of design temperatures in intermittently occupied rooms is to be provided.

All guestroom hot and chilled water supply pipes shall be thermally insulated from the service duct branch to the room including the fan coil unit.

For administration offices cased type fan coil units with supply air from plant and mechanical extraction to be used.

PUA to have air handling units with heating/cooling coils and variable frequency drive. Air distribution through ceiling diffusers supplied via ductwork.

Electrical motors for fans, hot and chilled water circulation pumps to comply with Eco-design Directive (2009/125/EC) which includes EuP (Energy Using Products) and ErP (Energy Related Products) and must meet the energy efficiency index (EEI) of less than or equal to 0.27 and IE3 quality standards. All fans and pumps over 5 kW(e) must be fitted with a variable frequency drive (VfD or VsD).

Earthing – bonding: In the mechanical/technical rooms care shall be taken to evaluate the need of earth bonding of the AHUs, cable management systems (cable basket trays, racks, etc.), incoming water steel pipes etc.

02.02. Plantroom/Shafts

In the design of plantrooms for equipment installation care is to be taken in ensuring sufficient space is allowed for easy access for maintenance and replacement, with particular care in respect of any heavy items of equipment that may require machine access for lifting in/out of position.

Ducts, riser shafts, etc. to be provided with easy access for maintenance and cleaning. An access duct accessible from within the corridor is to be provided to all guestrooms for disconnecting and/or regulating the hot & cold-water supply connections and other service valves. Install access panels where the equipment is hidden to provide access for maintenance. Avoid placement of access panels in lobby areas and in restaurants. In case of hidden/not accessible equipment the final requirement for servicing to be agreed with RHG.

All access ducts to be lockable with approved “bit key” lock mechanism with universal key to allow access across property.

02.03. Design Temperature Levels

Heating and cooling systems shall be designed to provide the following indoor design temperatures: (Local country climate will need to be considered when finalizing the below temperature ranges: -

Area	Summer	Winter
Guestrooms	22°C (40-50% humidity)	22°C (25-30% humidity)
Public areas	24°C (40-50% humidity)	22°C (25-30% humidity)
Back of House areas	24°C	22°C
IT / Server room	18-22°C	18-22°C
- All Computer Equipment Rooms / IDF	21°C (40-50% humidity)	21°C
Elevator equipment room	24°C (40-50% humidity)	21°C
Kitchen	24°C (50-60% humidity)	22°C
- Prep Areas (Poultry, Meat, Fish)	22°C (50-60% humidity)	22°C
- Pastry	22°C (50-60% humidity)	22°C
- Ice Carving	20°C (50-60% humidity)	20°C
- Wet Garbage	18°C (50-60% humidity)	18°C
Laundry	24°C (50-60% humidity)	22°C

Provide humidity control in compliance with ASHRAE Standard 62.1 -2019 (Ventilation for acceptable indoor air quality) through the dew point temperature.

02.04. District Cooling

It is mandatory to use District Cooling where available otherwise chilled water to be generated using electrically driven air-cooled high efficiency chillers as further detailed below. Design to include all necessary spaces, heat exchangers, valves, pumps, control panels etc.

Main digital metering must be provided at the main primary connection line for proper monitoring and consumption analysis.

All circulation pumps should have one spare each sized to handle full capacity.

HVAC chilled/cold water production requires 24 hours per day availability throughout all seasons.

02.05. Centralized Chillers for Space Cooling

Typical chilled water system design practice to be based on either a constant flow/variable temperature difference or variable flow/constant temperature difference concept.

Systems can also be classified according to the levels of pumping. Systems that can be considered include:

- Variable primary flow
- Constant primary-variable secondary flow
- Variable primary-variable secondary flow

All chillers shall comply with minimum energy performance standards as per the EU Eco-design Directive (Directive 2009/125/EC) which sets out the minimum efficiency performance standards (MEPS) for air cooled, comfort chillers used for space cooling.

Chillers selection shall be based on the Seasonal Space Cooling Energy Efficiency (SSCEE) factor shown below which supersedes other energy efficiency indices. All air-cooled comfort chillers must have the minimum ratings shown in the table below:

Air Cooled Comfort Chillers Rated Capacity	Global Warming Potential (GWP) > 150	Global Warming Potential (GWP) < 150
<400kW	149%	134%
≥400kW	161%	145%

Investment in highly efficiency, low Global Warming Potential (GWP) chiller technology will reduce the overall CO2 emissions, reduce operating costs and improve public perception of a "green responsible" business playing an increasing role in purchasing decisions.

Minimum 2 chiller units connected in parallel shall be piped to allow delta T control during part load operation. Individually both chillers at a ratio of 80% of their total maximum efficiency must be able to provide for the maximum demand capacity.

Heating produced by chillers to be recycled for pre-warming sanitary hot water.

Primary chiller pumps and Condenser fans shall have VSDs

In-line Flow and Return temperature sensors to be installed to chilled water supply and associated secondary distribution pumps to be VSD controlled to optimize delta T via BMS.

A VRF SYSTEM can be provided for low mid-scale projects, final approval is subject to the technical team. Heating produced by VRF to be recycled for pre-warming sanitary hot water.

A chiller management system (CMS) must be provided to control and monitor the mechanical and electrical services in the chiller plant and will comprise field mounted data gathering panels, sensors and actuators.

Information on the system status, alterations to setpoints or schedules, and alarm announcement must be available via software application that must operate on a typical workstation PC.

The CMS can be part of the property's BMS or can be a separate system.

02.06. Hot Water Generation / Space Heating

Space heating systems must be centralized systems. The system must be a variable temperature – variable flow LTHW / LPHW (Low Temperature Hot Water / Low Pressure Hot Water) unvented system comprising central heat generating plant and a two-pipe flow and return closed loop circulation system, terminating at locally thermostatically controlled heat emitters.

In properties and spaces where air conditioning is provided, the FCUs / AHUs should also be considered for space heating purposes, subject to satisfying energy efficiency and capital install cost reconciliation.

Direct electrical space heating must not be used unless the heating season is very short or property is a resort comprising unconnected villas spread over a large area and the provision of a centralized system is therefore considered unfeasible.

02.06.1. District Heating/Boilers

District Heating must be used as the primary source of HVAC heating wherever available. Otherwise, use condensing boilers with 90% efficiency or biomass boilers with 85% efficiency.

Provide minimum 3 boilers one 100% cold standby, with the other 2 units run in cascade temperature control. The boilers at a ratio of 40% of their total maximum capacity must be able to provide the maximum demand capacity.

HVAC hot water production required 24 hours per day with availability throughout all seasons.

For all circulation pumps, one spare pump is required to ensure continuous supply in the event of a failure of a pump.

Steam Boiler (TDS < 3000PPM/ TH of makeup water line < 5PPM/Steam Pressure as per system design/Condensate Tank Temperature to be above 85 degrees)

02.07. Bathrooms

The heating of the bathrooms should be designed together with guest room heating/cooling. Under floor heating mandatory in extreme conditions in terms of overall comfort must be minimum 90x60cm adjacent vanity shower linked and controlled individually in each bathroom by the BMS as per the corresponding technical description about the room functions.

For certain locations with extremely cold climate conditions, i.e., ski resorts, additional heating system, i.e., heated towel rail, radiators, underfloor heating, etc. to be considered for guestrooms toilets to ensure guest comfort is maintained all as agreed with RHG D&T.

03. VENTILATION

03.01. General

All indoor units should be supplied with conditioned fresh air for mixing with room air. Each guest room to have a local, automatically or manually controlled isolating duct damper with a minimum 20% open setting for ventilation control during non-occupancy periods with all extraction via ducted toilet outlet which is correctly sized for room volume and velocity to achieve low noise level as set out below. or to constantly achieve one air change per hour rate.

Provide enough ventilation to all areas of the hotel in accordance with the occupancy densities and percentage fresh air volumes as specified by RHG D&T. Supply and extract to operate continuously.

Energy exchange between all central fresh air and extract air systems at a minimum, and other ventilation systems where feasible, must be provided. (Heat Wheel, Horseshoe Coil etc.).

Provide separate air handling units for fresh air and extract to guest rooms, corridors, BOH, restaurants, bar, board rooms, meeting rooms, ballrooms, kitchens and gyms, each individually monitored by a central control system. Some of the units to these areas may be linked but only as agreed with RHG D&T.

Zone air handling units by type of occupancy, solar orientation and time of operation to allow shut down or Eco mode of equipment when heating and cooling is not required.

Demand Controlled Ventilation and use of CO₂ sensors for control of fresh air supply to variably occupied function areas must be provided. Use of CO and smoke sensors must be provided to control car park ventilation. Variable speed kitchen extract systems controlled by smoke and temperature sensors located in the exhaust hood must be considered.

Proper balancing and correct calibration of all sensors and controls is essential for a successful energy efficient installation. Building envelope should be designed under positive pressure scenario however Kitchen, Laundry, Lockers, Toilets, Boiler Rooms, etc. are only areas to be designed under negative pressure condition.

For aesthetic reasons channel fancoils with linear diffuser grilles are to be used in PUAs unless technically not possible.

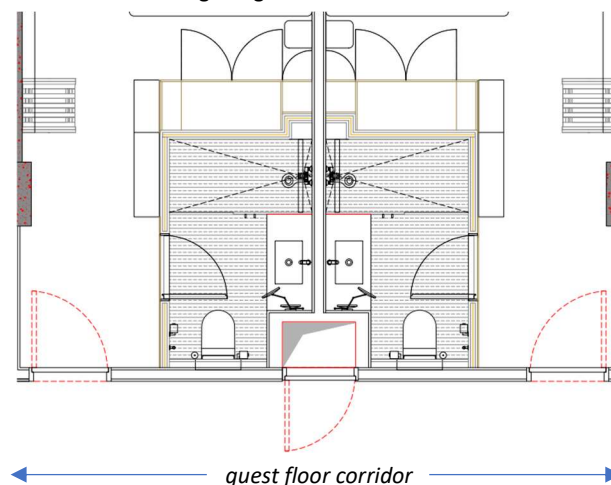
Earthing – bonding: In the mechanical/technical rooms care shall be taken to evaluate the need of earth bonding of the AHUs, cable management systems (cable basket trays, racks, etc.), incoming water steel pipes etc.

03.02. Plantroom/Shafts

In the design of plantrooms for equipment installation care is to be taken to ensure sufficient space is allowed for easy access for maintenance and replacement with particular care regarding any heavy items of equipment that may require machine access for lifting in/out of position.

It is not allowed to lay pipelines for household sewers in shafts with ventilation ducts.

It is good practice to design one shaft for two standard rooms, however for other types of the rooms an allocation of the shaft should be agreed with RHG. Access to the shaft should be realized from the corridor thus the technical staff can switch off the water supply, heating etc. without entering the guest room.



Ducts, riser shafts, etc. to be provided with easy access for maintenance and cleaning. An access duct accessible from within the corridor is to be provided to all guestrooms for disconnecting and/or regulating the hot & cold-water supply connections and other service valves.

All access ducts to be lockable with an approved bit lock mechanism all on single key type.

Outdoor ductwork is not to be used but when unavoidable provide accessibility (i.e., roof-mounted systems) and insulation (i.e., thermal, UV-resistant weatherproof outdoor insulation).

03.03. Fresh Air Supply & Extraction

The Mechanical Ventilation system is the most efficient and reliable system in the hotel and a filtered and treated fresh air intake system to be considered in all hotel areas as per international codes and RHG guidelines. Building pressure distribution and ventilation are mandatory for fire, life, safety, and health of the occupants. Fresh air with extract ensuring proper ventilation will improve indoor air quality by diluting polluted or stale indoor air and in addition pressurizes the area to help keep pollutants out.

Pressure distribution along hotel areas and functions also to be part of the ventilation design as per codes.

Air Mixing Plenums (mixing boxes) should be used in twin duct ventilation systems in order to increase heating air flow rates.

Provide anti-freeze protection control to protect coils from freezing by shutting down fans and closing outside air dampers when temperature at coils drops to between 3-4°C.

Provide motor operated dampers that automatically close when the fan is off.

03.03.1. Velocity

Air velocity to be controlled to limit any noise/vibration in the system. Coils must be selected for maximum air velocity of 2.5 m/sec. (cooling) and 4 m/sec. (heating).

03.03.2. Sound Level

Total noise level arising from all simultaneously operating mechanical and electrical plant as well as regenerated aerodynamic noise, whether it be duct borne, structurally transmitted or airborne, must be limited to the following levels:

Area	Sound Level
Guestrooms	Maximum sound level 35dB/A (ISO 3743) (32dB/A at medium fan speed) Bedroom night-time: L(A10)=30db
Public areas	40 dB/A in public areas (37dB/A at medium fan speed)
Back of House	40dB/A

Area	Sound Transmission Criteria Rw
Guestrooms Dry Walls / Partitions	Rw = 50 dB
Dry Walls between Guestrooms and Corridor	Rw = 48 dB
Ballroom / Meeting Rooms Partition (Movable or Fixed walls)	Rw = 42 dB

Different acoustic products and elements are to be considered for achieving sound levels:

- **Acoustic Lining:** In the discharge and intake duct of each low-pressure air handling unit, In guestroom fan coil ducts, In the discharge of each ceiling mounted FCU.
- **Duct Silencers:** Prefabricated low loss type duct silencers must be used where excessive noise problems are anticipated. Engineered sound attenuators must be used in high-pressure systems in central locations near fan discharge or inlet necessary to achieve maximum room sound levels. Splitter attenuators must be designed for smooth airflow and must be fitted for rigidity.
- **Acoustic Insulation:** Insulation material must be properly selected and specified. Chilled water pipes foam glass insulation is acceptable with aluminum cladding protection.

03.04. Guestrooms

Wall mounted digital thermostat controller to all guestrooms with following controls:

On/Off; with ECO mode for when rooms are unoccupied; Fan Low/Medium/High; Temperature minimum up/down $\pm 3^{\circ}\text{C}$. Digital display to be easy to use with a finish complimentary to wall finishes. Preferred solution with built in occupancy sensor. The thermostat height to be 1.40m to allow guests to easily read the temperature.

Location preferred on bathroom wall facing the guest room but subject to final design criteria.

Provide ventilation of enclosed areas such as bathrooms, dressing area, separated WC, etc.

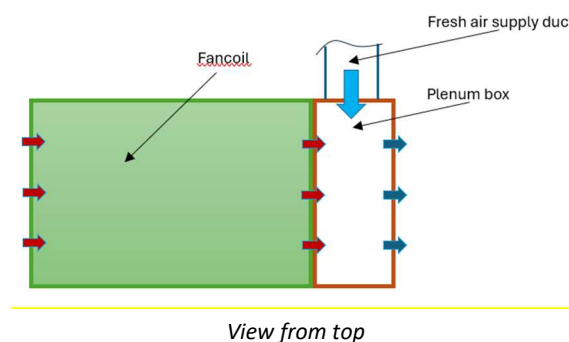
Occupancy sensors are the preferred option; however, an electronic key card controller will be accepted and is required in each guest room linked to air conditioning to switch off or step down the room temperature (ECO mode) when the room is unoccupied. In addition, a window switch is required for all openable windows to switch off the fan of the unit when the window is opened. Room controller to be linked to BMS and key card to the PMS or equivalent hotel management system.

Air Handling Units to provide ducted filtered fresh air which, depending on season, shall be either heated or cooled ($19/23^{\circ}\text{C}$), minimum capacity $85\text{m}^3/\text{h}$ per guestroom based on 2-person occupancy per room. This allowance should be increased pro-rata for larger occupancy family rooms.

Two types of fancoils are recommended to be used in hotels:

- fancoil units with extra plug for the fresh air duct connection where fresh air and circulated air are mixing inside the fancoil unit
- fancoil without such plug, but where the fresh air duct is connected to the plenum box.

For the second type the mixing plenum box should be used to mix air that comes from Fan coil and AHU for such purposes fresh air duct should be connected to the mixing plenum box from side in perpendicular air flow from fancoil.



In certain locations with hot climate, typically Resort location, in addition to HVAC system additional ventilation ceiling Fan in the room with 3 speed control need to be provided. Link to key card energy saver only not to window/patio door switch.

03.05. Bathrooms Extract

Air transfer from room through 30mm gap under bathroom door with extract grille over the toilet connected via ducted system with central fan unit, extract typically minimum $85\text{m}^3/\text{hour}$ based on 2-person occupancy per room. This allowance should be increased pro-rata for larger occupancy family rooms.

Noise transmission between adjoining bathrooms must be prevented through careful design of the ductwork and the exhaust system must contain baffles to prevent sight or sound transmission between bathrooms.

03.06. Public Areas

PUA to be equipped with dual fan systems (supply and extract). all according to European Norm 13779, that will allow partial (up to 70%) recirculation of ambient air monitored by CO_2 sensors.

The following fresh air volumes shall be supplied based on maximum occupancy:

- Meeting rooms: $43\text{m}^3/\text{hr}/\text{guest}$
- Multipurpose room: $35\text{m}^3/\text{hr}/\text{guest}$
- Restaurants and Bars: $50\text{m}^3/\text{hr}/\text{guest}$
- Fitness/ spa/ swimming pool: $60\text{m}^3/\text{hr}/\text{guest}$
- Reception/Lobby: $80\text{m}^3/\text{hr}/\text{guest}$
- Guest floors (corridors): $30\text{m}^3/\text{hr}/\text{m}^2$
- BOH: $43\text{m}^3/\text{hr}/\text{person}$
- Storage: $10\text{m}^3/\text{hr}/\text{person}$

An individual thermostatic control with temperature control only is required in each M&E room and room subdivisions, board rooms, private dining rooms, etc. all as agreed D&T RHG with air flows to FOH areas monitored by a Demand Control

Ventilation system (DCV) using CO₂ level sensing. DCV included as from the beginning of the engineering design, all connected to or part of the BMS System.

For aesthetical reasons channel fancoils and linear diffuser grills are preferred unless technical requirements are not allowed. Corridors and circulation areas should have a minimum of 30% air volume renewal each hour and air to be cooled or heated to ensure that difference of temperature between rooms and circulation areas does not exceed +/- 2°C.

Car and lift shaft ventilation according to the EN81-20 and EN81-50. Ventilation shall ensure the equipment, electric cables, etc., are protected from dust, harmful fumes and humidity.

03.07. Car Parking

To be fully ventilated by means of mechanically assisted ventilation system linked to a detector ensuring that carbon monoxide levels do not exceed safe limits and to be connected to BMS.

For all indoor/Underground areas, provide ventilation system for supply/extract of air interfaced to CO/CO₂ detection system to ensure levels do not exceed limits as required by local/international standards. There can be static or mechanical system provided no safety risk is posed or alternative jet fans can be used.

Provide a direct interface with fire alarm system for full speed operation of system in case of evacuation.

03.08. Back of House

All stores and linen rooms to be provided with mechanical air extraction with air handling units for the laundry extract equipped with special filters to both extract and supply.

Team members toilets/changing room areas to be provided with a dual fan system, extract and supply, according to European Norm 13779 with air volume as per MEP matrix. In general, in BOH, a volume of 43m³/hr/person to be applied and storage areas to be ventilated based on 10m³/hr/m².

All BOH Administration offices should have wall mounted thermostat controllers in each room with temperature control only.

03.09. Service apartments & presidential suits

A separate exhaust system should be designed for all ovens or induction panels.

03.10. Kitchen & Laundry Extract

03.10.1. Kitchen

Provide an induction type extract canopy with minimum 30cm overhang connected to a roof mounted extract fan all compliant with local codes and with enough extraction air volume required by type and extent of kitchen equipment within the kitchen. Ventilated ceiling type extract is also permitted. Kitchen extraction must not be connected to the comfort ventilation ductwork. The system must comply with the NFPA 96.

The 100% kitchen hood exhaust air volume to be replenished with conditioned make up air of which 65% is fresh air, 25% ambient kitchen air and 10% from surrounding areas.

High-efficient capture jet with built in fan at hood, less than 5% of the exhaust air volume, direct from kitchen air.

The hood extract fans to be located in a suitable position to disperse exhaust away from building and prevent any re-entry of contaminated air into outside air intakes or discharge onto adjoining terraces, pool decks, etc.

The use of UV filters in conjunction with primary grease removal device with cyclonic type filter is recommended and will be a requirement when the fan is located more than 30m from the extract hood or where cleaning access is difficult.

Suitable access to be provided to kitchen exhaust ducts to allow for economic cleaning at regular intervals.

Hoods to be equipped with a fire suppression system (i.e., Ansul R102 type or similar approved). Direct interface with fire alarm system to be included to shut off fans in case of fire evacuation.

Kitchen hood's exhaust fan to be VSD controlled and make up air supply for kitchen must be monitored via a Demand Controlled Ventilation (DCV) system connected to temperature and smoke (laser) sensor in the hood and control dampers at multi hoods to regulate air flows following the use of kitchen equipment. Kitchen hood intelligent controller to be connected to BMS for monitoring and control.

A separate exhaust hood shall be installed for the dishwashers connected to a moist air type duct.

Provide sufficient conditioned supply air to ensure temperatures within the kitchen remain below 26°C comfort levels with sufficient supply air diffusers to avoid draft. Supply air outlets should not be too close to the kitchen hood to avoid interference with hood performance.

03.10.2. Laundry

All dryer machines and flatwork irons (Calanders) should be equipped with separate exhaust ventilation. It is mandatory that all extraction ducts from dryers are provided with anti-pile filters with easy access for regular cleaning.

03.11. Emergency Staircase

All enclosed emergency exits must be provided with a pressure fan to maintain positive pressure within these areas. No cooling or heating facility is generally required in this area, however in temperate climates heating may be required.

03.12. Fire Life and Security

Air conditioning systems and ventilation systems for indoor air quality to provide automatic cut off in the event of activation of the fire alarm. All interfaces between Fire alarm and HVAC equipment to be directly connected through IM (Interface Modules) and not through any other systems such as BMS, or similar.

03.13. Scent technology

All PUA to have scent technology using RHG approved equipment to introduce scent into the ventilation system all as agreed with RHG D&T. For Park Inn and Prizehotel brands, scent technology is optional.

03.14. Air Filtration

Filters must be provided up with a stream of cooling/ heating coils in the fresh air and return air systems.

Filters must be located in sections of duct systems where there is little or no air turbulence to allow even distribution of air over the filter face. Maximum air velocity through the filters must not exceed 2.5m/s or the figure recommended by the manufacturer, whichever the lower. Filter banks must be made up from the filter manufacturer's standard units, rigidly braced within a galvanized sheet steel structure.

Filter housing must be provided having airtight access panels for filter replacement, complete with corrosion-resistant filter frames and clamps, gasketed and sealed to prevent air bypassing.

At the time of handover of the building the whole filter installation must be in "as-new" condition.

A large variety of filters are available offering different levels of efficiency, installation costs, replacement cost, and maintenance costs. The selection of filters must therefore be based on an evaluation of preferred indoor air quality, costs, local atmospheric conditions, and the capability of the maintenance personnel.

Minimum EU7-type filters must be used. In difficult conditions, pre-filters are to be used.

Different Type of Filters to be considered:

- **Washable Type (Aluminum Filter):** zinc electroplated steel mesh screen panel type and minimum 25mm thickness.
- **Disposable Low Efficiency Type:** disposable media panel type of 50mm thick with a filter media of continuous filament fiberglass or synthetic fibers.
- **Disposable High Efficiency Type (Box Filter):** fiber glass, disposable cartridge type 150mm thick. Filter cell sides must be constructed of galvanized steel.
- **Disposable High Efficiency Type (Bag Filter):** filters must consist of micro fine fiberglass or synthetic media, galvanized steel header and pockets retainers.
- **Carbon Filter:** must be made of moisture and grease resistant material. Activated carbon must be granular virgin coconut-shell base. The carbon weight must be 3.5kg/m³.
- **Grease Filter:** The grease filter must be light weight mesh type. The all-metal construction of the filter must be rust and corrosion resistant.

A differential pressure gauge must be provided for each filter bank for air handling units and the gauge must be fixed in such a position, outside the casings that it is accessible and easily read.

On completion of the works and before opening of the hotel all device filters must be replaced, including all guestrooms.

03.15.Pool areas

Comfort ventilation shall not be mixed with the swimming pool ventilation. Fresh air volumes as in section 2.5. Pool air temperature should be 2 to 4°C higher than pool water temperature (Indoor pool water temperature between 28 and 32°C). Proper dehumidification shall be performed in winter/summer mode. Provide factory applied corrosion protection to all AHUs and ductwork. The corrosion class in the choice of the ductwork, heat exchangers, fans etc. shall be reviewed and agreed with RHG D&T. In general, galvanized steel and powder coated aluminum are the recommended materials. Provide air to windows and to skylights to prevent condensation.

03.16.Air curtains

Recirculating or non-recirculating air curtains should be installed in the main entrances in order to create a barrier inhibiting the heat transfer from one zone to another. The air stream should be preferably not heated, and the door opening must be completely covered by the stream of air generated by the air curtain with airflow controlled and adapted to the outdoor climate. The heating source for the curtains can be electricity or hot water, from the main heating system. Due to the aesthetic reasons, it is preferable to use electrical curtains instead of hot water sourced for all PUAs and the appearance and color should match the design of the premise.

03.17.Refuse Area

Provide ventilation extraction to the refuse room with wet waste/or compactor. In addition, odor ozone machines should be installed.

03.18.Chiller/AHU/VRF Coil Protection

Coils to be treated with an anti-corrosion coating to avoid premature failure.

04. HOT/COLD WATER

04.01. General

All plumbing and drainage to be installed in accordance with the local standards and code of practice together with due regard to future maintenance and as such shall be planned to provide safe and ergonomic access requirements.

Supply hot and cold water to all service areas within the hotel together with drainage with sealed floor drains in all wet areas or areas subject to spillage or requiring floors to be wet washed.

04.02. Plantroom/Shafts

In the design of plantrooms care is to be taken in ensuring sufficient space is allowed for easy access for installation, maintenance and replacement of all equipment with particular care in respect of any heavy items of equipment that may require machine access for lifting in/out of position.

Ducts, riser shafts, etc. to be provided with easy access for maintenance and cleaning. An access duct accessible from within the corridor is to be provided to all guestrooms to allow for isolation and/or regulating the hot & cold-water supply connections and other service valves. Pipework shall be positioned within the riser so as not to restrict access to the riser or other services.

All access ducts to be lockable with approved "bit key" lock mechanism with universal key to allow access across property.

All penetrations through fire walls shall be fire stopped and appropriately certified and recorded in the register.

04.03. Legionella and Water treatment

A major concern to all hotel operators is the risk of legionella and all necessary steps are to be taken within the design process to eliminate any risks of this occurring including eliminating all dead-end situations and ensuring minimum/maximize temperature ranges are maintained throughout the entire supply system. Special care is to be taken with cooling towers which may be present within existing hotels.

Ensure necessary water treatment systems are provided subject to local requirements and water quality, such as filters, water softening, chlorination, legionella treatment, etc. The water system is to be designed to allow periodical tanks disinfection without water supply to the hotel, cleaning and legionella sampling for laboratory testing.

On sites where water hardness is extremely high and water softening requires excessive salt bulk salt saturators shall be installed to reduce manual handling requirements.

Multi-media pre-filtering system must be installed in all hotels with the addition of chlorine dosing system for the main tanks where the system has historic issues with uncontrollable bacteria growth. Chemical dosing systems (anti-corrosive/anti-scale) to be fitted to all boilers, steam generators and domestic water supply systems. All domestic water should pass UV-treatment.

An extra water filters (such as Big Blue 10 micron) are required to be installed directly in the laundry room if there is a risk of rusted water. These types of filters are recommended as good practice as well for high category rooms such as Presidential sweet etc.

Selection of pipework materials must be compatible with the method of dosing to ensure that the installation is not adversely impacted by the chemicals being used.

The installation is to be connected to the BMS where the hot and cold-water temperatures can be monitored, and trend logged for compliance. As a minimum the hot water storage cylinders, flow pipework from plate heat exchangers, HWS return pipework in the plantroom and cold-water storage tanks shall be monitored.

04.04. Cold Water Storage/Supply Distribution

04.04.1. Cold Water Storage

The reliability and quality of the available local water supply (municipal and/or borehole) will greatly influence the potable water system design features. In locations where the municipal water supply is classed as drinking water standard, the hotel may apply a simple filtration system to the incoming supply together with chlorination/ UV sterilization where necessary and/or where significant quantities of water are stored for long periods all as agreed with RHG D&T.

If an on-site purification and filtration system is required, the design and construction requirements are to be confirmed with RHG D&T. A small reverse osmosis purification plant may be necessary for limited applications such as food services areas and ice machines.

Domestic Water (TH (<60PPM) / TDS (<400PPM) / PH (7.2-7.4)/ Chlorine Level (0.3 to 0.6PPM at farthest point)

If the water supply exceeds 7⁰TH grains hardness, the entire hotel water system supply shall be softened. If laundry service is to be provided, then water supply shall be softened to maintain 1 to 3 ⁰TH grains hardness. Use of borehole water will in most instances require significant treatment using a reverse osmosis system.

On-site cold-water storage capacity shall be designed to sustain periods of unavailability for operation of the hotel in addition to fire protection requirements that will be established after consultation with the local fire authority. System to be design so cleaning and disinfection requirements do not affect the system providing ongoing uninterrupted supply to the hotel.

As a general guide allow the following for central cold-water storage:

- Minimum of 200 liters/guestroom per day in countries where the water supply is reliable and can be restored within 12 hours. To be taken into consideration that during the peak load hours The coefficient of simultaneity can reach 70%.
- In countries where water supply is unreliable or not guaranteed, it's allowed a minimum storage/generation capacity of 400 liters/guestroom per day and increase to 600 liters/guestroom per day with full laundry facilities.
- For resorts, it's allowed 800 liters/guestroom with two days storage. This additional capacity is to cover guestrooms consumption and all operational requirements, i.e., cleaning of all interior and exterior areas of the resort, all outlets, etc. (for resort hotels with landscaped gardens, swimming pools and spas recycling of grey water must be installed to minimize the storage and avoid use of freshwater usage for irrigation, toilets etc.).
- For water stressed countries, allow a minimum of 2-days water supply capacity with final agreed storage capacity subject to detailed study and agreement with RHG D&T.
- All water storage tanks should have thermal insulation and means of maintaining the stored water below 20°C.
- All tanks shall be fitted with high-level and low-level sensors connected to the BMS.
- For maintenance purposed bypass line for all water reservoirs is to be designed.

04.04.2. EU Water Supply Directive

All water supply must fulfil minimum the latest EU directive 98/83/EC of 3rd Nov. 1998 as amended by regulations 1882/2003/EC and 596/2009/EC or any subsequent amendments and installation must include: -

- The design of the water network will be done for eliminating any risk of legionella, including ensuring no dead-end stagnant water situations occur.
- Water treatment measures as required depending on quality of water based on analysis (i.e., may require multimedia carbon removal):
 - Automatic chlorine dosing system for domestic water tank to maintain chlorine levels according to international standard, for legionella prevention.
 - UV lamp to be installed on the cold water supply line to main kitchen for vegetables, and food washing.
 - Water Softeners should be installed, depending on the water quality.
- Pre-cooling of incoming water supply if temperature exceeds 20°C,
- Separate water lines to guest floors and FOH/BOH areas, designed and sized according to these standards.
- Pipework to be of non-corrosive, non-toxic material,
- Pipework with higher insulation coefficient to be provided in long length water networks to reduce losses,
- Anti-hammering measures on all pipe runs,
- All pipes clearly labelled for easy identification,
- Water booster with a minimum of 3 duty pumps each rated at 50% of maximum demand ensuring continuous supply at adequate pressure for both hot and cold-water installations,
- Separate electronic meters and isolation valves to be provided to main water intake including sub-meters in all major usage areas, i.e., kitchen, spa, landscaping, etc. Separate electronic meters also be installed for areas where sewage charges can be avoided, i.e., irrigation systems, etc.,
- Redundant pipework, pipework stubs that are not in use shall not exceed two pipe diameters from the main pipeline. On pipework 100mm diameter and above should not exceed 150mm,
- Boreholes to be utilized wherever possible with certification to ensure fit for human consumption,
- Grey water recycling systems are to be installed for landscaping irrigation sprinklers/ponds, water features, car washing, laundries, etc. and is to be further considered for toilet flush. For resort hotels a grey water recycling plant must be installed to further boost grey water catchment,
- Rainwater must be collected and used for landscaping irrigation sprinklers/ponds, water features and toilet flushing,
- Cold water storage shall be separated into raw water and treated/filtered water tanks,
- Back flow preventers installed to separate potable/non-potable water,
- Internal walls of the tanks where cold water is stored shall be washable, hygienic surfaces, i.e., fully tiled or surface coating which can be easily cleaned/disinfected.

04.04.3. Water Temperature

Water temperature in all piping before mixers to be below 20°C or above 50°C at all times and maximum 60°C, with required distribution temperature in guest rooms minimum 55°C after 60 seconds.

Where local regulations dictate for DDA or ambulant rooms, temperatures may need to be blended to provide 43°C to basins, baths and showers.

04.04.4. Water Pressure

The water pressure at the point of usage should be a minimum 2.5 and maximum 3.5 bar, except for the kitchen where the pressure should be minimum 3.5 bar. Pressure should not exceed 5 bar.

Pressure reducing valves for water should be installed locally in the shafts, directly in front of guest rooms for hotels above 6 floors. Such reducing valves should provide clear pressure indication or separate manometers should be installed.

Pressure to be balanced between hot and cold at all points of usage.

04.04.5. Water Flow Rates & Controls

Flow regulators must be installed to control flow rates as described in this section.

Flow rates at hand or rain showers to be 9 liters per minute with thermostatic mixer and fitted with flow regulators rather than flow restrictors. Shower heads to be the latest design that minimizes water usage but retains the feel-good factor.

Flow rates at wash basins hot and cold taps to be 4 to 6 liters per minute with thermostatic mixer.

Flow rates for hygienic showers should be 5 liters per minute or less with thermostat set point temperature 37-38°C.

Flow rates at kitchen sink hot and cold taps to be 10 liters per minute.

Toilet cistern to have a dual saving flush plate with maximum consumption of 6 liters / 3 liters per flush, dual flush water efficiency label A. Photocell activated controls to be installed to all washbasins and urinals in PUAs and BOH areas together with timed shower controls to all leisure and team members facilities. Spray type mousseur to be fitted to pots and crockery washing facilities within kitchens.

04.05. Hot Water Generation/Storage/Distribution

Use of District heating/cooling if available is mandatory. The availability of this shall be established at the start of the design process.

Solar heating/heat pumps, high efficiency boiler or condensation boiler to be used with efficiency rating of 90% or better for condensing boilers and 85% or better for biomass boilers. Where the cost ratio between electricity and gas/oil is less than 3 hot water heating use of heat pumps to be used as this will be operationally more economical than installing oil/gas boilers.

Hot Water must be continuously available through the heat exchanger with hot water storage vessel sized to cope with peak hours of usage. Where direct gas fired water heaters, plate heat exchangers or indirect cylinders with integral coils are used, a minimum of 2 no. units shall be provided and configured as N+1 where N is capable of providing the maximum demand needs for the hotel.

Design consumption shall be 150 liters per guestroom per day at 60°C plus 5m³ per day for the kitchen.

For business hotels the hot water demand should be designed to be capable of delivering all the hot water requirements over a 60-minute peak period. For leisure hotels this can be planned around a 90-minute peak period.

For resorts hot water temperature to be designed to be 65°C with smart mixing valves at entrance of each block, villas, bungalow, chalet, or resort building. In addition, individual Solar panels with electrical backup per block or building whenever the weather conditions allow.

Kitchen and laundry hot water requirements to be assessed and agreed following the equipment design and selection.

All hot water storage tanks must be tested for legionella bacteria on a regular basis and the tank heating to be capable of raising stored water temperature to over 60°C for a minimum of 1 hours per day or as recommended by local test laboratory and/or regulations.

The maximum waiting time for hot water to reach temperature in guest bathroom is 10 seconds, and in all other areas 20 seconds and achieve a minimum of 55°C in 60 seconds. Careful design attention must be paid to secondary return pipe sizes and pump capacities, especially for high rise buildings.

Storage tanks, distribution pipes, valves, etc. shall be thermally insulated with the optimal thickness of insulation as required for the size of pipe.

Distribution & secondary return pumps to be fitted with High Efficiency motors and VSD's. All pipes to be clearly labelled for easy identification. The main branches of the hot water return pipework serving the bedroom risers shall be fitted with thermic balancing valves which automatically regulate and balance the return water volume to maintain the circulation to achieve the desired 50°C without the need of manual regulation.

04.06. Kitchen Water Supply

Water supply for equipment to have necessary softening treatment with hot and cold supply throughout kitchen and support areas. i.e., dishwashers, combi-ovens, etc.

All kitchens are required to have a valid water certificate as required by local code but as a minimum to be in line with the requirements of Council Directive 98/83/EC of the European Union November 3rd, 1998, for “Quality water intended for human consumption”.

05. PLUMBING

05.01. Plantroom/Shafts

In the design of plantrooms for equipment installation care is to be taken in ensuring sufficient space is allowed for easy access for maintenance and replacement with particular care in respect of any heavy items of equipment that may require machine access for lifting in/out of position.

Riser shafts, etc. to be provided with easy access for maintenance and cleaning. An access duct accessible from within the corridor is to be provided to all guestrooms for disconnecting and/or regulating the hot & cold-water supply connections and other service valves.

All access ducts to be lockable with approved "bit key" lock mechanism with universal key for ease of access across property.

All pipework that penetrates through fire compartmentation walls shall be suitably fire stopped at floor, ceilings and walls. If the fire strategy does not require the riser to have a fire break on the floor and ceiling level a suitable work platform must be provided for access.

Piping material to comply with international standards.

Earthing – bonding: In the mechanical/technical rooms care shall be taken to evaluate the need of earth bonding of the AHUs, cable management systems (cable basket trays, racks, etc.), incoming water steel pipes etc.

05.02. Drainage Systems

Ensure adequate drainage is provided in all BOH wet areas, external circulation, parking, etc. All drainage to be in polyethylene high-density (PEHD). Where required to be provided sewage treatment plants are to be located at the lowest geographical location on the site, all designed for ease of maintenance and must be designed to allow for use of treated effluent for irrigation.

Provide rodding eye access at bottom of each vertical stack, collector ends, all directional changes, at end of each horizontal drain and at minimum 6m in straight lines.

Oil Interceptor is to be provided for large parking areas.

Drains above suspended ceilings to be cast iron with mechanical couplings and anti-condensation insulation, with acoustic insulation if above PUAs and to be fire stopped wherever they penetrate through floor slabs or any fire compartmentation barrier. Drainpipes within guestrooms to be carefully positioned and acoustically treated and where possible drains within bedroom area of room to be avoided.

Roof design, gutters and placement of downpipes to be carefully designed to ensure water drainage is away from the building and controlled to avoid washing over or onto walkways.

Site drainage to be designed to prevent standing water on the site except where engineered detention/retention ponds have been designed as part of the landscaping, or if required by local regulations.

Typically for remote resort locations, the Sewage Treatment Plant (STP) is an essential requirement for irrigation of the landscape areas, golf courses, etc. Location of STP plant to be coordinated with architect as per site layout, located away from all public and guestrooms areas.

05.02.1. Kitchen

Trapped floor drains should be provided in all wet areas complete with heavy-duty stainless-steel grating, floor slope to allow for a natural flow into the drain. Floor drains should be fitted to avoid creating any slip, trip or fall risk and should be suitable for trolley traffic. All covers must be removable for cleaning and maintenance.

All kitchen drainage must be connected to a grease interceptor. A soap/detergent interceptor should also be installed if laundry is located within the hotel. The Hot water supply should be provided to all interceptors to facilitate cleaning.

All interceptors located in an area where guests will not be affected during the cleaning process with external connection provided for direct collection via effluent tanker.

05.02.2. Bathroom

All hot and cold-water supply pipes should have an individual ball valve for each item of sanitaryware accessed from within the bathroom for local isolation. The main isolation valve for the whole bedroom shall be located in the service riser accessed from the corridor. Distribution and location of isolating valves and chambers to be provided along the various buildings, blocks, chalets etc. to allow isolation of the different areas. Hot water system must provide instant hot water with sufficient water pressure required for adjustable thermostatic spray shower without temperature fluctuation. All tap fittings to be high quality, water saving type with ceramic disc cartridge flow control with anti-scald features all labelled with internationally recognized symbols.

An automatic basin tab valve model in all BoH staff toilets to be provided.

Where possible, pipework should be avoided at all times. If pipework cannot be concealed, it must be electroplated copper with a chrome finish.

Floors drain with non-return device and stainless/nickel bronze steel grid to be located next to bathtub with sufficient slope. Use combined water trap with basin is needed to keep trap wet and therefore avoid smells. Access for cleaning the trap shall be from within the bathroom.

Shower facilities shall be provided with a water trap incorporating a non-return device. Access for cleaning the trap shall be from within the bathroom.

All drainage grilles, shower heads, tap aerators, etc. to be easily removable to allow for regular cleaning requirements for legionella prevention.

05.02.3. Sanitary Ware

Sanitary ware shall be as per the Brand Standard and constructed of good quality porcelain, generally white or as per interior design and securely mounted in position. Special care to provide support on non-structural walls, i.e., dry wall partitions, etc.

05.03. Kitchen

05.03.1. Compressor (Heat rejection from Cold rooms)

Where equipment requirements require installation of refrigeration condenser external to the building ensure suitable access for maintenance is provided. Condensation pipework must be connected to a suitable drain.

05.04. Gas Supply

Incoming gas supply sufficiently sized to cover 110% of the maximum demand.

All gas ducts to be clearly marked using the national color code.

The main gas supply valve shall automatically close in the event of a fire alarm in addition to a manual shut-off valve. Each kitchen and all other areas where gas is consumed must have a clearly marked and easy to operate emergency shut-off valve.

Gas meter to be installed on the main incoming gas supply with sub-meters at each area where gas is consumed and interfaced with BMS.

Appropriate gas detection alarms to be installed in all areas where any gas is consumed or where risk of leaks exist, CO, CO₂, etc.

06. BMS (Building Management System)

06.01. General

The BMS system will be run over a comprehensive structured cabling network that covers the entire hotel including all FOH and BOH areas, typically consisting of multimode fiber optical cabling for the backbone with minimum CAT6a cabling for horizontal distribution. Acceptable communication protocols at the field, network and management levels are: Modbus, BACnet, KNX, Dali (lighting control), M-bus and TCP/IP.

06.02. Building Management System

A fully automatic, programmable electronic BMS complete with the latest software version, and which can be easily updated via the internet shall control, operate and monitor the mechanical and electrical installations. BMS interface should support national and English languages. For software interfaces at the BMS management level OPC standards should be used. BMS system should be interfaced with the PMS system to receive information about reservation for guest rooms and meeting rooms (see also software section below). The BMS system must be capable of integration with the chosen hotel management system (e.g., EMMA). The system shall include access to the BMS system locally (mobile control stations with possibility of manual adjustment and control) or through the remote system at the central cloud service. Provide daily back up locally on the central servers.

The following MEPs shall be connected to and controlled by the BMS: -

- Boilers and all associated temperature sensors,
- AHUs including heat recovery and all associated temperature sensors,
- All fans & pumps VSD's,
- Fans/Filters/Dampers,
- Chillers/Chilled water circulation pumps and all associated temperature sensors,
- Room Fan Coil Air Conditioning units (FCUs) and all associated temperature sensors,
- Lighting/Dimming interior and exterior,
- Fire Alarm Systems (for monitoring purpose),
- Sprinklers, and/or fire suppression system (IT rooms).
- Elevators,
- Cold rooms temperature and alarm monitoring,
- All main and sub-Metering/Energy Management.
- Pressurized units and all associated alarms
- Generator
- AC units in server room (or other technical rooms, lift machine room, etc.)

The controls functionality of the BMS as a minimum standard shall comply with EN 15232 Class B¹ (Energy performance of buildings - Impact of Building Automation, Controls and Building Management) for the following parts: -

- Heating control
- Cooling control
- Ventilation and Air conditioning control
- Lighting control; Occupancy control
- Technical Building Management

The HVAC mechanical systems must be designed to support the above-mentioned BMS functionality, i.e., they must allow for variable flow and demand control strategies with appropriately set up duct and piping networks (e.g., parallel supply/return air ducts, hydraulic circuits for variable flow, etc.) and to be equipped with the necessary components (e.g., VAV-boxes, VSDs, etc.). At the design stage, the BMS supplier shall provide a comprehensive list for approval with a description of the function of the monitoring and control points tabulated as Analogue and Digital Inputs & Outputs. The BMS system is to be interfaced with the PABX to forward all alarms notices and with the fire alarm system in order to shut off all necessary devices should the alarm be activated.

06.03. Hardware

- Working station requirements:
 - **Operating system:** Approved/supported versions of Microsoft server editions
 - **Management platform:** Hpe ILO advanced or similar
 - **Product family:** HPe Proliant DL380 Gen 11 or above or similar

¹ *"EN 15232 Class A" is preferred where feasible, due to further reduced energy consumption impact and these sections being applicable:

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- **Disk redundancy:** Minimum 4 disks
 - **Raid set-up:** Minimum RAID5, RAID6 or above
 - **Memory Requirements:** Minimum 16Gb
 - **CPU requirements:** Minimum 2 CPUs
 - **Power requirements:** Redundant power supply
 - **Monitor:** 27"
- All the components shall be in compliance with Directive 73/23/EEC (Low voltage equipment) and Directive 89/336/EEC (Electromagnetic compatibility). Indoor equipment to be operational in ambient temperature range +5°C-45°C and relative humidity 20%-80%. All the equipment shall be CE marked. Outdoor equipment to be protected from the UV exposure, wet weather and frost.
 - All BMS equipment is to be enclosed appropriately to provide environmental, electrical and mechanical protection and security against unauthorized access. All equipment enclosures are to be labelled using a long-life robust labelling system. Key information and key documentation are to be kept in each enclosure where the equipment details can be found. On the panel front, Hand/Off/Auto switches to be provided for all the AHUs. A BMS service switch is to be provided in order to prevent the alarm generation whilst maintenance work is being carried out. When the BMS panel incorporates a gas valve, a gas valve reset switch should be fitted to the panel front.
 - All controllers must be open protocol, IP addressable and programmable. All components must be clearly labelled with their LAN reference and IP address. Packaged controls must be supplied with a description of operation and control strategy drawings. They shall be interfaced with the BMS by the BMS contractor who will be responsible for ensuring that the packaged controls integrate seamlessly with the BMS and that monitoring, and control points can be viewed and adjusted directly from the BMS supervisor.
 - Cabling: Distance between BMS low voltage cabling and power cabling to avoid electromagnetic interference and communication disruptions shall be considered as per the corresponding regulations. Provide labelling to both ends of the cables. All the cabling installation practices being complied with EN 50086-1.
 - Sensors: Provide temperature sensors, PT100 or PT1000, with an accuracy at least class B, IEC751. Avoid installing temperature sensors close to heat sources or air flows that may influence readings. Outside air temperature sensors must be mounted in shaded north facing locations. Always install temperature sensors in the district heating system in order to monitor the difference temperature between the incoming and the return pipe system. Accuracy required is $\pm 1\%$ of reading. In-line Flow and Return temperature sensors to be installed to chilled water supply and associated secondary distribution pumps to be VSD controlled to optimize delta T via BMS. Provide absolute or differential pressure sensors for chilled water and condenser water applications measuring accuracy at least $\pm 0.5\%$ FS. Provide pressure sensors according to the designed pressure range with field signaling 4-20mA or 0-10V. Provide flow sensors with field signaling 4-20mA or 0-10V. For VAV sensors can also use 2-10V signaling.
 - Damper motors (actuators): Provide with open/close indication. Preferable voltage power 24V, alternatively 230V. Damper motors, installed in the intake supply will be designed to work to minimum ambient temperatures -30°C. Provide test motioning (self-motion) for the fire rated damper motors through the BMS on a weekly basis.
 - All meters to be connected to the BMS (BACnet, M-bus or Modbus communication protocols) and be accessed remotely. Provide two-way communication between the meter and external networks (both supply & grid) for maintenance and control of the meter. The following features shall be provided by the BMS in regards with the monitoring of the meters: Monitoring and recording (hourly, daily, weekly, annually), compare consumption for the recorded periods for trend analysis, determine "out of range"/outliers values and alert building maintenance operator, record peak energy consumption.

As a minimum the BMS shall report the status of the following systems:

1. Chiller
- Run Status
- Chiller Start / Stop Command
- Indication of manual switch position
- Common Alarm /Trip
- Chiller water leaving temperature
- Common header water leaving temperature
- Common header water entering temperature
- Compressor status
- Butterfly valve open/close commands and status
- Common header chiller supply flow switch
2. Primary chilled water pump
- Start/Stop command
- Pump run status
- Pump switch position Hand/Off/Auto
- Pump alarm/Trip status
- Duty/Standby auto changeover
3. Secondary chilled water pumps
- Start/Stop command
- Pump run status
- Pump switch position Hand/Off/Auto
- Pump alarm/Trip status
- Pump discharge pressure at common header
- VFD trip alarm
- VFD enable/disable command
- Duty/Standby auto changeover
4. Chiller pressurization unit
- System pressure
- Power supply status
- Alarm/trip status
5. Air Handling Unit (AHU)
- Temperature and humidity of intake air from outside
- Cooling coil entering temperature
- Cooling coil leaving temperature and humidity
- Air supply temperature and humidity after heater
- Fan start/stop command
- Fan run status
- Filter dirty status (Pressure difference)
- Supply air smoke damper
- Fire signal from the alarm panel
- Fan trip alarm
- Fan switch position Hand/Off/Auto
- Cooling coil command
- Electric heater control signal
- Freeze stat alarm (frost protection)
- Duty/Standby auto changeover of Fan
6. Fan coil unit (Public Areas)
- Fan speed command
- Fan run status
- Monitoring space temperature
- Cooling coil command

7. Staircase pressurization fan
- Fan run status
- Air inlet damper command
- Damper status
- Fan switch position Hand/Off/Auto
- Fan trip alarm status (Low and High speed)
- Breaker on/off status
- Staircase pressure
- Fan start/stop command
8. Lobby pressurization fan
- Fan run status
- Air inlet damper command
- Damper status
- Fan switch position Hand/Off/Auto
- Fan trip alarm status (Low and High speed)
- Breaker on/off status
- Fan start/stop command
9. Car park extraction fan
- Fan run status
- Fan switch position Hand/Off/Auto
- Fan trip alarm status (Low and High speed)
- Breaker on/off status
- Fan start/stop command
- CO level status
10. Car park fresh air
- Fan run status
- Fan switch position Hand/Off/Auto
- Fan trip alarm status (Low and High speed)
- Fan start/stop command
- Filter dirty status (Pressure difference)
11. Smoke extract fan
- Fan run status
- Fan switch position Hand/Off/Auto
- Fan trip alarm status (Low and High speed)
- Fan start/stop command
12. Sewage Pump
- Pump run status
- Pump start/stop command
- Pump alarm/trip status
- Pump switch position Hand/Off/Auto
- Water level indicator from water tank
13. Filtration pump
- Pump run status
- Pump start/stop command
- Pump alarm/trip status
- Pump switch position Hand/Off/Auto
14. Swimming pool pumps
- Pump run status
- Pump start/stop command
- Pump alarm/trip status
- Pump switch position Hand/Off/Auto
- Supply and return water temperature

15. Heat exchanger
- Inlet temperature (Primary line)
- Outlet temperature (Primary line)
- Inlet temperature (Secondary line)
- Outlet temperature (Secondary line)
- Cooling coil command
- Heat recovery - Low efficiency ratio alarm
16. Heat exchanger swimming pool
- Inlet temperature (Primary line)
- Outlet temperature (Primary line)
- Inlet temperature (Secondary line)
- Outlet temperature (Secondary line)
- Cooling coil command
17. Calorifiers
- Hot water calorifier temperature
- Unit on/off command
- Unit alarm/trip status
- Unit switch position Hand/Off/Auto
- Unit run status
- Flow Temperature
- Return Temperature
18. Booster pump set
- Pump run status
- Pump start/stop command
- Pump alarm/trip status
- Pump switch position Hand/Off/Auto
- System output pressure
- VFD alarm/trip status
19. Water Tanks
- Level indication
- High- and Low-level alarm
- Water temperature for roof top cooled tanks
- Water level for fire for basement water tanks
20. Transfer pump set
- Pump run status
- Pump start/stop command
- Pump alarm/trip status
- Pump switch position Hand/Off/Auto
- System output pressure
21. Sump pumps
- Pump run status
- Pump start/stop command
- Pump alarm/trip status
- Pump switch position Hand/Off/Auto
- High- and low-level alarm of PIT
22. Hot water circulation pump (Flow and Return)
- Pump run status
- Pump start/stop command
- Pump alarm/trip status
- Pump switch position Hand/Off/Auto
- No water flow status
23. Fire pump set
- Jockey pump run status
- Electrical pump run status
- Diesel pump run status
- Jockey alarm/trip status

-	Electrical alarm/trip status
-	Diesel alarm/trip status
24. Lift	
-	Power supply status for lift control panel
-	Common alarm
-	Cabin call alarm
25. LV Main switch gear	
-	On/Off status of main incoming breakers
-	Trip alarm of main incomer
-	Earth fault alarm of incomer
26. Generators	
-	Status generator breaker on/off
-	Generator status MCCB trip status
-	Generator common alarm indication
-	Generator not in auto mode alarm
-	Generator voltage and frequency
-	Fuel level in daily tank
-	Status of fuel transfer pump
-	Generator battery status
27. CO detection system	
-	System on/off status
-	System alarm status
28. Cold rooms and freezer rooms	
-	System on/off status
-	Temperature reading
-	Alarm/trip status
29. Boilers	
-	System on/off status
-	Temperature/steam pressure reading
-	Alarm/trip status
-	Daily fuel tank level
-	Status of fuel transfer pump
-	Flow Temperature
-	Return Temperature
30. Boiler Plant room Supply/Extract Fans	
-	Fan run status
-	Fan switch position Hand/Off/Auto
-	Fan trip alarm status (Low and High speed)
-	Fan start/stop command
31. Boiler Flue Dilution Fans	
-	Fan run status
-	Fan switch position Hand/Off/Auto
-	Fan trip alarm status (Low and High speed)
-	Fan start/stop command
32. Kitchen hood fans	
-	Fan run status
-	Fan switch position Hand/Off/Auto
-	Fan trip/alarm status
-	Fan start/stop command
-	Fan VFD run status (if any)
-	Fan VFD alarm trip (if any)
33. Facade/Outdoor lighting	
-	Control through time schedule and lighting sensor
-	Run status On/Off/Time schedule
-	Time clock function/ lighting sensor outdoor lighting
-	Time clock function/ lighting sensor external signage

-	Time clock function Enter
34.	Smoke Hatches
-	Status open/close
-	Alarm status
35.	Variable Speed Drives (VSD)
-	VSD status/fault status
-	VSD speed (%)
36.	Chlorine Gas Alarm (Pool Plantroom)
-	System on/off status
-	System alarm status

Provide individual selection of the delay times for all the above alarms.

06.04.Automation Controls Air Handling Units

- Fans to be controlled from PLCs through time schedule. Variable Air Ventilation units that are servicing local areas with important differences in occupancy (Demand controlled ventilation), there shall be additional control through CO₂ sensors.
- The BMS system shall be interfaced with the Fire detection system. The latter shall automatically close down ventilation plants according to the building fire strategy and agreement with the fire authorities. On activation of the building fire alarm system, a signal will be sent to each control panel in order to close the comfort ventilation. The test-motion of the fire dampers shall be automatically done at least once per week according to the corresponding time schedule.
- Provide Air Handling Units with battery frost protection.
- Consider night-time back cooling during the spring/summer season in the atrium areas.
- Heating recovery - thermal efficiency ratio to be viewed to the BMS interface. Alarm to be generated at low efficiency ratio.
- Compensation and optimization program techniques shall be implemented as an aid to energy conservation.

06.05.Electrical

- Provide monitor all major circuit breakers status, trip and phase fail relays.
- Electricity meters to use BACnet or Modbus communication protocol.

06.06.Guestroom BMS controlling.

06.06.1. System functionality

The system shall incorporate intelligent control at each guestroom, controlling heating/ cooling (fan coil unit operation, radiators) with a linked system controlling the main plant. Wall mounted digital thermostat controller to all guestrooms with following controls: On/Off; with ECO mode for when rooms are unoccupied; Fan Low/Medium/High; Temperature minimum up/down +/-3°C. Digital display to be easy to use with a finish complimentary to wall finishes.

Preferred solutions for controlling occupancy are sensors. There should be enough Occupancy Sensors in the room to ensure the functionality, from built in Occupancy Sensors in the thermostats and independent Occupancy sensors, to Door contactors/sensors that allow monitoring the entrance/exit cycles. The system will be reliable for controlling occupancy when guests are sleeping and when several guests are in the room, and maybe in different locations (i.e. one sleeping and the other one in the bathroom...). In case sensors cannot be implemented, alternatively an electronic key card controller can be accepted. In both cases, occupancy sensors or key card controller, it is required they are linked in each guest room to air conditioning to switch off or step down the fancoil units, based on setting the room temperature depending on the different options of occupancy, as below detailed.

It should be provided separate temperature setpoints for occupied/non occupied mode, check in status (pre-comfort mode), irrespective of the design parameters (temperature and humidity) used for sizing the equipment. In addition, a window switch is required for all openable windows to switch off the fan of the unit when the window is opened. Room controller to be linked to BMS and key card to the PMS or equivalent hotel management system. Occupancy to be linked to BMS and to be controlled through the check in signal from the PMS. Provide both means of communication between the BMS and the room controllers.

The BMS should be connected to the hotel PMS in order to receive information about the status of the rooms (checked in, checked out, out of order).

The different modes that should be available in the room controllers, where the setpoints can be adjusted by the hotel engineering team:

Comfort mode: guest checked in and occupancy sensors or Keycard switch activated:

- When the occupancy sensors or keycard switch are activated (keycard inserted in the keycard reader), then the system goes into Comfort mode. Setpoint temperature in Comfort mode: 20C/23C winter/summer (recommended – editable in the BMS for the engineering team). The guest will be able to adjust the above setpoint +/-3C.
- Fancoil to be activated into the Comfort mode. (The guest can adjust the fancoil in 3 steps manually (override the automatic function))
- Heating floor to be activated into the Comfort mode.
- The damper for supply air and exhaust goes to predefined maximum designed air volume (applicable only with VAV systems).
- Lights and sockets linked to occupancy (not the 24h ones) will be activated.

In case of digital door lock system is chosen in the hotel, the abovementioned functionalities need to be implemented with alternative equipment, so that the energy is fully optimized.

Pre-Comfort mode: guest checked in and occupancy sensors or Keycard switch deactivated:

- When the occupancy sensors or keycard switch are deactivated (keycard not inserted in the keycard reader), then the system goes into the Pre-Comfort mode. Setpoint temperature in Pre-Comfort mode: 19C/24C winter/summer (recommended – editable in the BMS for the engineering team).
- Fancoil will be automatically supplying the required temperatures, running/stopping depending on them.
- Heating floor to be activated.
- The damper for supply air and exhaust goes to predefined maximum designed air volume (applicable only with VAV systems).
- Lights and sockets linked to occupancy (not the 24h ones) will be deactivated.

In case of digital door lock system is chosen in the hotel, the abovementioned functionalities need to be implemented with alternative equipment, so that the energy is fully optimized.

Economy mode: guest checked out, irrespective of the situation of occupancy sensors or Keycard switch:

- When the guest is checked out in the PMS system, irrespective of the situation of the occupancy sensors or keycard switch, then the system goes into Economy mode after 45 minutes. Setpoint temperature in Economy mode: 19C/24C winter/summer (recommended – editable in the BMS).
- Next time that a guest (new or the existing) reactivates the keycard switch, the Comfort temperature setpoint to be considered (20C)
- Fancoil to be deactivated into the Economy mode.
- Heating floor to be deactivated into the Eco mode.
- The damper for supply air and exhaust goes to predefined minimum designed air volume. (applicable only with VAV systems).
- Lights and sockets linked to occupancy (not the 24h ones) will be deactivated. However, the system will have the functionality of activating the lights and sockets when the occupancy sensors/keycard activate (for Housekeeping, Maintenance...). The rest of the elements linked to the HVAC will remain according to the economy mode.

In case of digital door lock system is chosen in the hotel, the abovementioned functionalities need to be implemented with alternative equipment, so that the energy is fully optimized.

In case the connection with the PMS is not possible (D&T team will have to agree on this point), there would be only 2 modes (Comfort and Economy mode). Comfort mode will be considered when the room is occupied (occupancy sensors or Keycard switch activated) and Economy mode will be considered when the room is not occupied (occupancy sensors or Keycard switch deactivated).

Green mode

The system will be prepared for having a 4th mode. The green mode would allow the guests to opt in, to further energy savings. A one touch Green Mode button on the thermostat shall provide sustainably minded guests with an easy way to help conserve energy during their stay. Pressing the Green Mode button shall change from the Comfort mode to the Green mode, where the setpoint/ventilation in the room would be defined by the hotel (1C/2C less in winter mode or 1C/2C more in summer mode, and 1/2 fan speeds less). Optionally on the display, a message such as 'Green Mode - Thank you for helping to save energy' (or any other linked to RHG Responsible Business policy) would appear.

Out of order mode:

- All climatic systems are in sleep mode until the guest room temperature drops to 16°C, then the systems will start to heating up until it reaches 18°C or if the temperature will raise up to 27°C then fancoil will start colling until the temperature is reduced to 25°C.
- The damper for supply air and exhaust goes to predefined minimum designed air volume.
- Lights and sockets including the 24h ones will be deactivated.

In case of digital door lock system is chosen in the hotel, the abovementioned functionalities need to be implemented with alternative equipment, so that the energy is fully optimized.

06.06.2. Open Window Sensor

When the window contact registers the «open window», the fancoil stops, ventilation is reduced to the minimum air volume, and the regulator regulates the room according to a fixed frost limit of 12 ° C. If the temperature falls below this limit, the fancoil starts and the heating valve opens and then stops again when the room temperature reaches 15 ° C. In economy mode the fancoil is off but it should start running as soon as the temperature falls below the frost limit and until the room temperature reaches 15C.

06.06.3. Bathroom:

The heating floor in the bathroom is to be activated as per the pre-described modes (Comfort and Economy), so it will have to be linked to the keycard (or equivalent digital door lock system). The BMS will have the possibility of defining a -minimum temperature of the bathroom heating, for Economy mode.

The temperature for the heating floor in the bathroom should be regulated via a temperature sensor in the bathroom floor individually per room and is set via the SD system. The recommended temperature would be 24 ° C but can be adjusted by the technical staff in the hotels. The guests cannot change this value themselves.

06.06.4. Parameters in the BMS:

- Reading positions on Door contactor/sensors, room occupancy sensors or key card
- Reading position on window contact
- Room temperature readings
 - Comfort mode setpoint temperature
 - Pre-Comfort mode setpoint temperature
 - Economy mode setpoint temperature
 - Green mode setpoint temperature
 - Out of order mode
 - Real room temperature and Guest setpoint temperature
- Reading of bathroom floor temperature (Set the temperature in the bathroom floor)
- Damper reading position on supply air and exhaust (minimum or full air volume) (applicable only with VAV systems)
- Reading of the fancoil speed
- Reading of valve position heating valve
- Reading of valve position cooling valve

07. SMART Guestroom system

07.01. System overview

Hotels are increasingly demanding advanced levels of controllability over their facilities. As a result, a flexible control system, encompassing services such as lighting, temperature, audio visual (AV) and blind/curtain control is becoming an essential item in hotel guestrooms and suites, providing the ability to enhance the guest experience and increase comfort while minimizing energy wastage.

These standards intend to be a guideline for the implementation of reliable systems, allowing a wide range of functionalities to improve the guest comfort.

In this chapter we're referring only to the Room Management System. We had referred to it in the previous chapter referred to the BMS, but the level of functionality requested for the Smart Guestroom System is further beyond what was included in previous chapter, where only the FCU (fancoil units) and temperature modes requirements were detailedly explained. For this particular point, what we referred to in the latter chapter is fully applicable in this one, so the Occupancy modes, temperature setpoints, protection parameters... must be respected, in this one.

The manufacturer's product range must support both UL and CE certification so that the system architecture can be used globally. This requirement allows clients to have a global specification and define consistent features and functionality for all projects regardless of the region where the system is installed.

Suppliers for the smart rooms should have full Integration and networking gateways to other systems: Ethernet 10/100 Base T – KNX – BACnet interface (via Ethernet) – DMX512 Tx and Rx – Somfy via RS-485 – Modbus via RS-485 – RS-232 – Low level Dry Contact – Analog 1-10V input – Network real time clock. The following systems should be reliably integrated:

- PMS – Systems: EMMA, Opera,...
- Access Control / Doorlocks: Assa Abloy, Dorma Kaba,...
- Operations – Systems: integration with Doorlock systems including Amadeus HotSOS.
- APIs – Systems must provide a secure, profile-based APIs for other 3rd party systems integration.
- Messaging – Messaging from the system should be supported when creating accounts or for the sending of alerts and warnings.

Ethernet Devices will support TCP/UDP, IPv4/IPv6, unicast/multicast/broadcast, DyNet1, DyNet2, Fidelio and 'Text and Binary Integration' protocols. Device shall operate within at least 0° to 50°C and at 0% to 90% relative humidity non-condensing environment. The device shall supply power to the RS-485 bus, not exceeding 300 mA.

The system will be designed in accordance with CE compliance requirements, the network control system shall comply with the applicable European EMC Directive, Low Voltage Directive, Radio Equipment Directive, ROHS Directive, REACH Directive, Cybersecurity certification and other relevant international standards.

The system shall be installed according to the site-specific System Installation Guide supplied by the control system partner organization. The lighting control system shall be tested by the installer to meet all the requirements set down in the specification. The system shall be set up initially in accordance with the [clients and/or consultants] instructions and left working. The commissioning engineer shall perform acceptance testing and produce a snagging list of issues for the installer to rectify.

The system shall provide guests with immediate feedback to confirm their actions and any automatic room functions. To achieve this, the Room Control Units (RCU) must have independent intelligence, able to fully operate a room or suite of any size on a local basis. Failures at a server level may affect the dashboard and integrated services, however rooms shall continue to function.

In the event of a power failure, current guestroom settings shall be stored in non-volatile memory. So, when power is restored, the previous settings and user preferences are automatically restored. In the case of power loss, the RCU shall be able to fully restore the room state, including guest preferences such as lighting scene and temperature setpoint. Recovery of room state must be done immediately on power up, without any dependency on network or server access.

07.02. System architecture

The control system shall be a networked type, using the existing IP network, with either Ethernet or optical fiber cabling to the room. The control system shall provide distributed control, with user interfaces and sensors communicating directly with load controllers, without requiring the intervention of a central processing unit.

Gateways and bridge devices shall be available from the system manufacturer to enable the creation of a trunk and spur network topology. Individual network spurs shall be galvanically isolated from trunks, with devices that enable message filtering so as to minimize unnecessary trunk and spur network traffic. It shall be possible for controllers or gateways to connect to networks of luminaires, sensors, and user interfaces.

All network devices are to be configured from the same communications port as their operation port. A single communication port for both configuration and operation is required to ensure that configuration changes to the control system can be

performed quickly and easily without the need to use multiple protocols, changing the network cabling, or having direct access to a network device. This also allows for changes to the system from the head-end user software.

All load controllers and UIs shall be supplied with a basic configuration that allows the control system to operate in a default way immediately upon installation. This allows for UIs to send on, off and preset selection commands onto the network. All load controllers correctly connected to a room subnetwork shall respond to these messages appropriately. Achievement of this functionality shall not require any commissioning.

All configuration information saved to devices on the network shall be stored in the relevant device, contained within non-volatile EEPROM or FLASH memory. This data shall remain secure for an indefinite period, upon loss of supply.

Each Room Control Unit (RCU) shall have a direct Ethernet connection to the IP network. They will be sized according to the elements to be commanded (FCU, lighting circuits, outlets). Guestrooms, suites, and public areas such as restaurants and function rooms, shall have a permanent connection to the server. IPv4/IPv6 shall be selected in the project design phase for each site. Network gateway redundancy shall be provided by adding an alternative floor gateway for every primary floor gateway in the system. The two floor gateways shall be in physically different locations to ensure adequate redundancy. The primary gateway and the alternative gateway shall be both connected via an Ethernet Trunk Connection. If a floor gateway fails, then head-end software and the RCU connections will automatically switch over to the working gateway.

Sufficient IP addresses shall be made available for the control system with a minimum of one IP address required for each RCU, Ethernet gateway, head-end software server, and integrated systems.

The control system shall support:

- Server-to-server interface with PMS and access control systems.
- The building network must support end-to-end encryption and prevent any form of direct room-to-room communication.
- Devices should communicate via specialized guestroom technology open standards, with open compatibility to third party devices including, but not limited FCU's and light points.
- Profile based user security policies.
- Detailed device-by-device monitoring and control.
- Different profile types for different room types.
- Configurable default settings for different seasons.
- A unified dashboard to monitor and control all rooms, suites and public areas in real-time.
- The architecture should be scalable depending on the size of the property, up to 3000 rooms, suites and public areas.

Network communications must be hardwired inside the room and to each room. Each room control unit shall provide communication over TCP/IP to the control system server and third-party TCP/IP systems. Room to room looping is not acceptable to avoid latency and a single point of failure.

All load controllers must internally support the required resources such as mains power supply, micro-processor, memory, direct communication port, and internal logic so that devices are independently responsible for their functionality. This is required so that there are no hidden extra accessories that are required for network devices to operate. Control systems which are dependent on external network accessories for devices to achieve their core functionality will not be accepted.

The control network within rooms would preferably utilize an RS-485 multi-drop control bus, with data connections between devices on the network with four-conductor type (two conductors for distribution of a Safety Extra Low Voltage (SELV) DC supply only, to power keypads and sensors). Devices connections should preferably be pressure pad type screw terminals, RJ45 or RJ12 connectors. The control system manufacturer shall also offer a range of appropriate network gateways to connect network segments.

07.03. System functionality

The control system shall be designed to:

- Provide intuitive room control.
- Ensure vacant rooms are not consuming unnecessary energy.
- Allow energy consumption to be managed in an automated way.
- Enable service staff to be coordinated around room occupancy.
- Minimize maintenance costs and downtime.

The configuration software shall allow users to create a standard configuration template for most common hotel applications, such as guestroom, suite, bar, restaurant, lounge, meeting room, ballroom, spa & gym, retail etc.

The system shall work with the areas, scenes and room numbers for easy operation and commissioning.

The system shall have programmed responses to occupancy and room mode, where room facilities go to a particular state when guests are in the room and to other states when guests leave. Lights, A/C, window covering etc. will be on at a selected level at places where people are present and on a standby level when not present.

All room state values shall be configured to meet the hotel's preferences, either globally or by room profile. They shall be able to update automatically from day to night and from season to season and shall be self-managed by authorized hotel dashboard users.

07.03.1. Room Status & Guest Services

In a very short period, a guest needs to orientate themselves with different services in the room. Controls shall be intuitive so the guest can quickly learn to use them. From a wall panel, guests can control multiple functions such as "Welcome" or "Goodnight" lighting scenes, blinds/curtains, temperature, and entertainment from one button push. The control system shall be capable of performing multiple functions from a single input trigger. For example, when a guest first enters the room the system selects a welcome lighting scene, sets HVAC to occupied, open blinds, and reports the updated room status to the RCU.

Networked rooms provide many benefits including automatic reporting of guest service requests (room status) to the central head-end software. When the guest activates the Do Not Disturb (DND), Make Up Room (MUR), Laundry Pick Up (LPU), or Room Service Pickup (RSP) status, the status shall be synchronized with the dashboard and property management system.

The system shall use context from a range of sources to switch the room between the following room modes:

- Checked Out: Unoccupied or Occupied (Staff Mode)
- Checked In: Unoccupied or Occupied.
- For the latter (Checked In & Occupied) we need to control Guest Preferences (Daytime, Evening, Green Mode, VIP mode)
- Out of order

The sources that provide this context and knowledge shall be within the system, as well as through integration:

- Real-time Occupancy (Source: System Sensors/entrance switch). Motion sensors and/or entrance door switch, the system shall determine if a guest is present in the room.
- Real-time Clock (Source: System Network). The real-time clock shall be used by welcome scenes and wakeup alarms (where specified) to present guests with time-of-day appropriate lighting scenes and curtain logic. This feature shall be provided by a combination of server application and network gateways.
- Check-In/Out Events (Source: PMS Integration). Guestroom rented/sold status shall drive energy management. Receiving these status changes from the PMS shall be a key trigger to change guestroom mode and default settings.
- Occupancy Type - Guest/Staff (Source: Access Control Integration). Determining whether a guest or staff member is entering / inside a room shall help to both optimize room state, as well as preserve guest preferences throughout their stay. Knowing that a staff member rather than a guest has been in the room shall allow for shorter timeout periods when leaving the room to further aid energy savings.

A switch embedded in the entrance door frame shall control the door events and trigger the occupancy check sequence to start. It shall indicate that a guest is entering or leaving the room, and that occupancy needs to be checked. The occupancy state shall remain until triggered by another door open/close event and if occupied, the state shall remain until no movement is detected, or access card is removed, and a chosen timeout period has expired.

Occupancy sensors shall be multifunctional. Sensors shall contain a photoelectric (PE) sensor, a motion detector (PIR) and an IR receiver in the same package and support all three detection elements at the same time. Sensors shall be available that utilize either Passive Infra-red (PIR) or a combination of PIR and Ultrasonic (US) for occupancy detection. Occupancy sensors shall have a user-definable occupancy timeout. All sensor configuration settings shall be made from software via the control network.

In the event of a guest leaves the room whilst another is sleeping and is not detected, a recovery mode shall ensure that the remaining guest is undisturbed, silently recovering the previous state as soon as sensors detect movement.

07.03.4 Lighting

Switching

In the controller, switching functionality is performed by relays. When grouped into logical areas, combinations of lights can be switched simultaneously to create different lighting scenes. All output channels can be configured for switching only, if required. The electrical design should ensure that the number and types of loads in each switched circuit do not exceed the relay power rating of the connected output.

Dimming and colour control

Load controllers shall be available for controlling different dimmable load types via the following protocols: Leading edge/trailing edge power dimming; 1-10 V, DSI or DALI broadcast signal dimming; DALI addressable dimming (DALI 209); DMX512. Controllers shall be available from the manufacturer to allow DALI tuneable white lamps and DMX colour control lamps to be specified.

Scene control

The system shall recall a lighting scene or sequence when triggered manually by a button press or from the dashboard or automatically when the room changes between unoccupied/occupied by guest or staff.

Scenes shall be created for each area in the room. Default scenes shall be Bright, Relax, Work, Goodnight and Leaving. Additional scenes can be specified in the system design.

Where provided by the controller, the preset scenes shall be defined for lighting intensity and colour to. Specific channels can be excluded from a preset or set to switching only. Preset scene options shall be available for hotel guest services, such as evening turndown or VIP guest welcome to be further enhanced with lighting. Using positioned light fittings and hidden lighting scenes activated by staff, a gift or note can be highlighted on the bed for returning guests.

The preset scene range shall be offset at sunset and returned to normal at sunrise. Because of this schedule, scenes can be dynamic, taking into account natural daylight or time of day, in achieving your intended effect. Rather than having multiple buttons for the same area to recall different lighting scenes – for example in the bedroom, Morning, Afternoon, Evening and Night scenes can be recalled by a single button on a user interface programmed so that the right scene is recalled for that specific time of day. This allows for a simplified experience with fewer buttons required. Examples of this logic can also be found in bathrooms to allow a softer light level upon entry during the night, instead of traditional controls which would just turn the lighting on to 100%. The options of adaptive configurations are only limited by the imagination.

Anti-stumble lighting

Anti-stumble sensors shall detect when someone gets out of bed and returns to bed during the night. Bedside left, bedside right and bathroom nightlights shall create a lit pathway from the bedside to the bathroom at nighttime. The pathway shall only be illuminated near the floor so as not to disturb others and shall automatically turn off when the guest returns to bed after a predefined delay period.

Wake Up lighting

The system shall provide a wakeup alarm with lighting feature to mimic sunrise by simulating the dawn light intensity and colour before the set wake time. The Wakeup lighting Wake up lighting can reduce sleep inertia and improve well-being.

Emergency lighting

Each room shall have an emergency input capability allowing for emergency egress activation of room lighting. The emergency input overrides guest controls until the event clears.

07.03.5 HVAC Control

HVAC control should be designed according to the chapter 06. BMS (Building Management System). FCU configuration, thermostats with functionalities modes depending on the Occupancy status.

MultiPoint Readings

Every user interface panel in the guestroom (entrance, bedsides, desk, curtains etc.) shall include an embedded temperature sensor that can be used to provide temperature reading points. The RCU shall locally process an average weighted reading from multipoint readings and provide a truly representative temperature to the HVAC controller to enable more accurate heating and cooling adjustments.

Energy metering for each guestroom could be either notional (based on the systems energy demand, based on the running time) or via physical submeter.

VIP Mode

While smart room conditioning is optimal for energy saving and is seamless for most guests, VIP's can be treated a little differently. The control system shall support a VIP Mode to temporarily disable automated room state changes where the hotel determines that guest comfort or happiness could be disturbed. VIP mode shall be enabled from the dashboard, to

prevent the room state changing irrespective of real-time occupancy. This preference shall be retained until check-out or cancelled by the guest, wherein the room shall be returned to its standard automated behaviour.

07.03.6 Power & Mechanical Control

General Purpose Outlets

The system shall have controller outputs for controlling general purpose outlets (GPOs) within the room and turn off some outlets when the room is unoccupied.

Curtains & Blinds

Window covering controls must be fully interfaced with all lighting and energy management functions proposed in this document. Window covering circuits can be configured for lighting control if required.

Window coverings shall be automatically closed when the room is checked-out to preserve room assets.

The system shall be able to monitor the open/close position and control motorized window coverings such as curtains and blinds, and other motorized facilities such as a projector screen. Systems that cannot control motorized window coverings shall not be accepted.

Doorbell

The system shall have controller outputs for controlling a room doorbell. The doorbell shall have a setting to deactivate the doorbell if DND is activated. The doorbell shall have a configurable delay period to temporarily deactivate the doorbell after each push and a configurable maximum ring time to limit the sound due to excessive pushing.

TV and Audio-Visual Control

The control system shall use a power circuit and the TV's 'power on' function to turn on the Audio-Visual equipment/TV and display the welcome message.

Additional Sensors

Additional and custom sensor options can be implemented to capture the following data: Light Intensity, Humidity, Room Safe Door open/close, Water Leaks, Soil Moisture, FCU Filter, FCU Drip Tray

07.03.8 User Interface functions

Using networked user interfaces allows the user to interact with the system more intuitively to control different system functions. Networked user interfaces shall be available with LED indicators and/or a display to give the user feedback to confirm button operation and current system status. The user interfaces shall communicate directly to the control system using the core protocol.

Button graphics shall be instantly recognizable to an international audience. Text shall be avoided where the requirement is to be internationally recognizable.

Inputs from network user interfaces shall be triggered by a button press from the user. Button press inputs shall be configured in the commissioning software button editor for each user interface. The Buttons Editor shall enable selection of icons and text for room information and functions on the UI display.

There has to be a minimum of 2 user interphase panels, one located on a wall in the middle of the room, and fully accessible, and the other one on the bedside. However, there could be more such as, entry panel, bathroom panel, bedside left panel, or bedside right panel. The user interface shall have from two to six buttons, with an optional LCD display variant that will allow for system information to be communicated to the user. This device should match the same look, feel and finish of the standard button variant. The device must have a built-in proximity sensor and temperature sensor. The device must support proximity triggered back lighting to enable the user to find the user interface locations during nighttime conditions. The device shall be available in polycarbonate and metallic finishes. The device must support triggering multiple functions on controlled services, such as lights, ceiling fans, air- conditioning, curtains etc.

A user interface in the corridor shall be able to indicate Do Not Disturb/Make Up Room and shall allow to trigger a doorbell function that can flash the room lights or ring the doorbell when paired with a third-party door chime.

Multi-language support

The user interfaces with a display shall enable the guest to select a language from the language menu on the display. The user interface shall send out a Set Language message so the hotel dashboard will reflect the chosen language. The language can also be set from the Hotel dashboard to the user interface.

The FIAS protocol shall allow the PMS to set a guest language on check-in and reset the language on check-out. The language for the room shall be reflected in the hotel dashboard and on the user interfaces in the room. If no language is received from the PMS or it does not match an available language, the display shall continue to show the language menu.

Labelling & Engraving

User interfaces shall be designed using an online configurator for selecting the design including product variant, finish, color, icons, and text. User interfaces could support custom laser etching of text, icons, and braille on each button for intuitive guest operation.

In room wake up alarm settings.

The user interface with a display shall allow the user to set the wake-up alarm time, snooze the alarm and cancel the alarm.

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Interphase with Hotel Services

The system should allow the services to the guests to be controlled via the User interphase panels. When a guest requests a service like Making Up Room/Laundry Picking Up/ Room Service delivery/Engineering problem in the room, the system should allow to send a notification to the Hotel systems (PMS, Hotelkit,...) for requesting.

07.03.9 System software

The control system shall use advanced Windows based graphical programming to configure all network devices and define the behaviour and relationship between network devices. The software must be capable of configuring any device within the system. The software shall not be dependent on extra plugins or drivers to enable configuration of devices.

The system must provide head-end server management software.

The server management software should be connected to all control system devices in the building, allowing automatic reporting of the room status to the central server. This shall allow system messages to be captured, such as preset state, channel levels and room status indications.

The Software operation will not be linked to any additional license fee. The server management software shall have an open API to integrate with a variety of hotel systems, so that real-time data can be fed into other systems such as housekeeping and engineering systems.

The server management software shall automatically send notifications to appropriate staff members for any room or system alert.

07.03.10 Dashboard

The system should have the functionality of presenting all data retrieved from the system monitoring and control, into a dashboard interface. The Dashboard must provide a visual representation of the following aspects, including option for changing parameters:

- Hotel floors and rooms. It must provide the capability for monitoring performance, searching, troubleshooting and management of issues for multiple guest rooms.
- Real-time summary of any room.
- Possibility to activate or deactivate guest room automation.
- Possibility to view (and adjust) current temperature, temperature setpoint, HVAC fan speed, HVAC operating mode room occupancy, lighting scenes.
- Possibility to view (and adjust) window covering positions.
- The dashboard must provide current and historical data from all the variables monitored (Lighting scenes and modes, HVAC modes/temperatures, windows/curtains status, room occupied, hotel services requested).
- The dashboard must enable the user to monitor system alerts across all guest rooms from a single page with an indication of the type of issue, the device in question and the floor/room that is affected.
- The dashboard must provide default and custom reports for users to view information on energy usage, system activity and diagnostics. Reported notional energy includes lighting and HVAC energy. Lighting energy reports shall require lighting zone power to be provided in Watts. HVAC energy requires HVAC zone power in BTU/hr to be provided.

Access to the dashboard shall be via a domain using a HTTPS website address and user login. Authentication shall be either be provided by local server authentication or by single sign-on using LDAPS.

Dashboard User Management shall offer a range of configurable user profiles and permissions, allowing individuals the appropriate level of control and access while removing unnecessary or restricted information from the interface. It shall provide session settings and logging of user activity.

Where a user profile grants such permissions, controls to allow remote change of services shall be displayed. For users without permissions, these shall be hidden, or the screen presented in a read-only mode.

The dashboard administrator shall be able to restrict user access to different floors. This may be useful to enable visibility and control of specific areas only for the relevant tenant.

The dashboard administrator shall be able to add and configure permissions for API users. This shall be by way of an initial username and password and then automatically providing a session token.

08. ELECTRICAL

08.01.General

Electrical installation must comply with the latest EN standards, local regulations as well as RHG standards covering all conduits, sockets, jack points, switches, controls, etc. for all socket outlets, light fittings, structured cabling, IT, telephony, safety and security and AV equipment, etc. all in accordance with local regulations. System to be energy efficient requiring minimum maintenance with access for maintenance appropriately considered based on lifespan for re-lamping intervals.

All PUAs to be equipped with dimmable circuits with controls conveniently located within each area/zone.

Provide wall and/or floor outlets for guest usage within seating areas of the lounge, bar and restaurant.

Provide power outlets for servicing at not exceeding 15m apart, i.e., outlets available within 7.5m of any location with a minimum of 2no.13A outlets per room.

Generally, one double power outlet supplied via central UPS and two double electrical power outlets to be provided per team members workstation plus additional outlets for all printers and peripherals.

No cable size to be less than 2.5mm for power supplies with minimum 1.5mm for uncontained power supply connections from loose fittings into outlets. All wiring within the building shall be copper. The wiring distribution shall be four wire color coded grounded system. Non – continuous grounded systems, i.e., cable trays, shall have separate ground with <25 ohms to ground.

Main switchboard shall have breakers with a single main disconnect switch and GFI protection.

Provide a sub-distribution panel on each floor or wing (depending on the number of rooms per floor maybe more than one) together with fuse panel per guestroom conveniently located for access in entrance foyer or wardrobe of guestrooms. A combined fuse panel for maximum 2 guestrooms is also acceptable, but the panel cannot be located in a technical shaft with wet systems nor within the guestroom itself. Sub-distribution panel boards shall be circuit breaker type, and a suitably rated RCCD fitted on the incoming isolator.

Panel Boards:

- Separate panels for light and power.
- Panel boards shall be located where possible to be within 30.5 m (100') of its loads.
- All panel boards should be in secured rooms out of sight of the public view, on the same floor as the load it serves.
- Separate panels for all tenants

Allow safety protection with appropriately sized fuses throughout. Circuit breakers should have selectivity coordination according to the local standards. Guestroom must be protected with a minimum 30 mA ELCB (Earth Leak Circuit Breaker), bathrooms and wet rooms – such as wellness/pool area, kitchen, laundry, irrigation landscape etc. – in general are protected with 30 mA Circuit Breaker Note that higher protection standards may be required under the local code. Locate all the equipment in an interior weather protected space unless an exterior location is absolutely necessary. Do not provide equipment designed for interior use at exterior locations. IP class must be considered.

08.02.Incoming Electrical Supply

Electrical supply capacity to be provided based on 120 % of maximum demand load. Allow 10% spare capacity on each distribution panel and on the main switchboard to allow for future expansion. Circuit Breakers to be properly labelled, indicating the loading and service provision.

An automatic voltage stabilizer to be provided for incomer feeders, for countries facing power disruption.

An automatic Power Factor correction equipment must be provided to maintain the power factor at a level of at least 0.95.

Local energy policy and tariffs to be studied in depth and used as basis for establishing the most energy efficient and environmentally friendly system design.

Feeders shall be copper cable in cable trays and in conduit where in concrete slabs and masonry walls.

08.03. Emergency/Standby Backup Power

Except in rare locations where the hotel can be supported by two different networks, an emergency generator is mandatory.

In locations with stable supply, we need to size the backup systems (below indicated) to keep systems running during 12h. For areas with frequent power cuts, a standby generator must be installed that generates 100% of the demand and must be provided with a minimum fuel reserve of 48 hours of operations or more dependent on reliability and/or response of local supplier. For emerging markets, an additional backup generator to be provided with min 50% capacity of above.

In resorts/remote locations the reserve is to be extended to a minimum of 96 hours.

Diesel engine driven generator with synchronization and automatic transfer switches to activate generator upon loss of power. Generators to have a built-in AVR Power Switch.

Rated to provide power for all code required and critical loads. Automatic transfer switches shall be provided to activate the generator.

Provide centralized back up batteries in case of main supply/generator failure with one-hour fire resistant cabling to all illuminated exit signs.

08.03.1. Emergency Generator Backup Systems Connected

As a minimum emergency power is to be supplied to the following: -

- Fire lifts
- Lift homing device (i.e., lift travel to evacuation level and open doors in event of fire)
- Pressurization fans / smoke control system
- Smoke removal ventilation systems
- Smoke hatches
- Sprinkler and Hydrant pumps
- Sumps pumps
- Critical access control systems (e.g., access barriers)
- Emergency and exit lights.
- Public Announcement system
- Critical fridges and freezers
- Fire detection and extinguishing systems
- Telephone equipment
- All other lifesaving equipment and critical security systems
- Security and CCTV systems
- Computer room and data system, PMS, POS, call accounting, Fidelio and UPS system.
- Building Management System
- Stairs and Exit signs (Guest and Public areas).
- Switch gear room, Generator room.
- Engineer office.
- Computer Room, Telephone Room.
- Laundry 50%, Kitchen 50%.
- Elevator room.
- Guest room corridor 30%.
- Administration 30%.
- Reception desk 50%. Public Areas 30%.
- Security office.
- Banquet area 30%, Gym and health club 30%.
- Pool lighting.
- Public toilets.
- Team members canteen.

In addition, the following are to be considered and agreed with RHG D&T: -

- Cooling / Heating plant (depending on countries)
- Conference & meeting rooms
- Public areas lighting (including Health & Spa club)
- Server
- CWS and DHWS pumps (where they are installed in a basement location)

08.04. Uninterruptible Power Supply

Two Separate UPS to be provisioned in the project. One is to be dedicated purely to the Server Room power supply to all servers, computers and racks, in addition to the IDF rooms racks and switches. And another dedicated UPS for the hotel specific areas and sockets.

a. One Main UPS for:

- Computer room and servers.
- PABX and telephone system.

b. One Main UPS connection shall support these services:

- Fire alarm system.
- CCTV and Access control.
- BMS system.
- Point of sale.
- Gas control.
- Sockets at Reception and working desks as per IT Guidelines.

The UPS shall be capable of being operated in any of the following modes:

- **Normal mode:** During normal operation, the UPS shall be used to provide precise regulated and transient – free power to the critical load. The primary AC source shall be used to supply power to the rectifier and battery charger separately via an isolation transformer. The rectifier shall provide regulated DC power to support the inverter while the separate battery charge is maintaining the battery plant in fully charged condition. The inverter shall convert the DC power into regulated AC power for the load.
- **Emergency Mode:** Upon failure of the normal AC power input, input power for the inverter shall automatically be supplied from the battery plant for 15 minutes. When the AC power is restored, input power to the inverter shall automatically be supplied from the rectifier while the battery plant is being recharged by the battery charger. If the input power does not return, the UPS shall automatically shut itself down while sounding an audible alarm when the lower limit of the battery plant is reached.
- **Bypass Mode:** If UPS failure occurs, a static switch shall transfer the load to the AC main power within ¼ cycle. Retransfer of the load to the output of the UPS shall be manually accomplished. For maintenance furnish and install make before break manually initiated maintenance bypass switches to bypass and isolate the UPS system without interruption of the load.
- **The UPS** shall have built in protection against under voltage and over current, including voltage and current surges on the output caused by load transfer.

08.05. Lighting

08.05.1. Internal lighting

The appointed lighting designer shall design the interior and exterior lighting to enhance the architectural concept/features and design details of the entire hotel including the exterior facades and landscaping. A schematic lighting design principles reference will be issued by RHG D&T as part of the overall briefing in RHG R&C guide and/or RHG Social Spaces Guide. As part of the design process all maintenance issues are to be addressed in terms of access and lifespan.

Day lighting preferred in areas where applicable and achievable.

Lighting levels to be appropriate to the usage of the space with minimum average lighting levels in accordance with EN12464 as follows:

- | | |
|--------------------------|--|
| • Bedrooms | = 200 lux (400 lux at desk) |
| • Bathroom | = 300 lux (400 lux in front of mirror) |
| • Guestroom lobby | = 150 lux |
| • Reception | = 200 lux (300 lux at desktop) |
| • Bar | = 100 lux (300 lux above counter) |
| • Meeting/Ballroom Rooms | = 400 lux |
| • Restaurant | = 300 lux |
| • Public Toilets | = 150 lux |
| • Lift Landings | = 150 lux |
| • Kitchen | = 500 lux |
| • Offices / BOH | = 300 lux |
| • Corridors | = 80 - 120 lux |

Total installed power for lighting to be less than 3W/m²/100 lux.

Sufficient circuits are to be provided in each of the PUAs to allow for implementation of the lighting designers scheme. All PUAs to have a proprietary computerized lighting control system with possibility of minimum 5 pre-programmed scenes per room. In addition, automatic dimming is to be provided to all PUAs exposed to daylight. For smaller areas as a minimum each light fitting type and each functional area must be controlled by an individual dimmer.

Lighting to all PUAs, including guest corridors, to be LED type lights with minimum 50,000 hours lamp life with 70% lumen maintenance under 25°C with converter chosen from manufacturer offering 5-year warranty. No halogen or incandescent lights allowed, and attention must be paid to the color of the lighting (°K) with color precision maximum +75 Kelvin -2

McAdams. Energy saving type lamps may be permitted for specific lighting design features and this is to be agreed with RHG D&T. Within guestrooms 100% of lighting must be energy saving LED type. For all guest areas, including guestrooms, warm light will be used (2.700K). Exceptionally in these areas it could reach up to 3000K, but needs to be approved by RHG. In the BoH areas, we could have a range between 3.000K and 3.800K.

Guest rooms floors corridors to have 2/3 of the lighting controlled by strategically placed motion detectors to bring on lights only as needed with remaining 1/3 controlled by a switch at the panel. All switching to be also controlled through BMS. Lighting to corridors to be on two circuits, one permanent (monitored by light sensor if sufficient daylight in corridor) with minimum 25 lux on guestroom doors and 15 lux to the rest of the corridor. The second circuit to be on motion detector and on timer to provide minimum 80 lux and maximum 120 lux to entire corridor. Ends of the corridors must be well illuminated to positively terminate the vista with increased lighting level in lift lobbies and lighting to be used to enhance the artwork.

For the Restaurants, careful use of various zones of lighting such as overall mood lighting, table-top lighting and focal lighting to display areas such as buffets, drinks display, food arrangements, etc. together with feature light fittings Typically feature lighting at every 25m² with positions of lighting to allow flexibility for varied table layouts. Lighting type must enhance food display areas. The lighting controls shall allow for pre-set mood scenes across the day and during mealtimes.

08.05.2. External Lighting/Building Illumination

External lighting should be LEDs and controlled by photocells, and timer together with BMS.

Outdoor and BOH lighting to be controlled by motion sensors, together with timer or BMS.

The exterior lighting of the hotel must be carefully designed in a way that takes into consideration various operational criteria together with the architecture.

From a functional aspect all areas must be sufficiently lit, primarily along the walkways and close to entrances with all parking areas and landscaped areas appropriately lit. In addition, lighting must, wherever possible, enhance the architectural appearance of the building, with emphasis on its 3-dimensional form. The lighting of the main building facades must be clearly visible to approaching traffic but avoid any direct illumination of the windows.

The color composition of the building lighting is important, and this is to be carefully considered within the lighting scheme. Key circuits, such as a port cochere, that require different timing sequences are to be on separate circuits.

Lighting must be a combination of waterproof recessed spotlights together with decorative wall mounted lights providing atmosphere to the entry and entrance areas.

All exterior lighting design must allow easy access for cleaning and maintenance. The location of all light fittings should be carefully considered to protect them from any damage and must be located within planted areas wherever possible.

Only LEDs lamps will be used.

08.05.3 Tennis Court

Tennis court lighting for club play shall consist of twenty-two 8-foot 0 inch-long four lamp continuous LED lights, eleven on each side of the court. Lights to be mounted 5 meters (16 ft. 6 in) above the court and tilted at a 45-degree angle. The lights shall be suspended from four posts on each side. Each post shall have an 11 ft. davit arm. Post shall be constructed of aluminum, light in weight, capable of withstanding 100 miles per hour wind.

Fixture housing shall be 1/8-inch-thick extruded aluminum for structural support and cool ballast operation. The door assembly shall be of heavy gauge extruded aluminum frame with prismatic high impact lens. The reflector assembly shall be made of prefinished aluminum, finished in white baked enamel.

08.05.4 Squash Court

Squash courts should contain sufficient quantity with LED luminaries to provide an average illumination level of 300 lux. The lamps shall be protected by an impact resistant lens.

08.05.5 Swimming Pool Electrical Installation

The distribution panel serving the pool shall have an isolated ground bus connected to switch gear ground bus not connected to pool neutral. All indoor wet areas should have electrical safety ELCB with proper earthing and IP rate.

Pool lights shall be LED, 12 volts wet niche type with an individual shielding isolating transformer provided for each light. Transformers should be located as close as possible to lights to minimize the drop of voltage.

No electric outlets of over 24 volts shall be located within 3 meters of the pool perimeter unless such outlets are protected with earth leakage circuit breakers.

Any metal part of fixture or conduits in contact with pool water shall be brass or any other approved non-corrosion – resistant material.

Pool motors and controllers shall be connected to a circuit protected by a ground fault circuit interrupter.

08.05.6 Emergency Lighting

Emergency lights should have fire resistant cables fitted in all escape routes with illuminated LED exit signs to indicate the escape routes. Emergency lights should be operational for a minimum of one hour and provide a minimum of 5 lux in all escape routes. Emergency lighting in technical rooms must be minimum 15 lux. Lux levels will vary depending on the specific area, due to local regulations, so this must be taken into consideration.

A percentage of the general light fittings should be connected to an emergency power circuit to provide emergency lighting. If individual battery-operated light fittings are provided, these must comply with local regulations and shall be of decorative units to match the overall design scheme.

Guestrooms to be also equipped with emergency lighting fixture LED at room corridor entrance ceiling.

Lifts to be also equipped with emergency lighting.

In outdoor areas, emergency battery lights should be placed in exterior doors and paths to guide guests to safe areas.

08.05.7 Car Park

Lighting within car parks shall be LEDs to have 2/3 of the lighting controlled by strategically placed motion detectors to bring on lights only as needed with remaining 1/3 controlled by a switch at the panel. All the lighting is also to be controlled by timer/connected to the BMS.

Lighting levels to be appropriate to the usage of the space with minimum average lighting levels in accordance with EN12464 as follows:

- Car park = 10 lux

Lighting of the exterior car park to required minimum levels with all areas of shadow eliminated. Within underground parking to provide LED ceiling fittings.

Provide waterproof emergency lighting with backup power with high and low marker lights in accordance with applicable regulations to all underground parking areas.

08.06. Guestrooms

RHG Rooms & Corridors Guidelines by brands have detailed information for room circuits wiring diagrams, sockets, switchers and provide information for the following set up.

08.06.1. Light Switching

Light switching is to be simple and easy for the guest to operate and to be designed to suit the interior design scheme. In accessible rooms light switches are to be located between 800 mm to 1300 mm above floor level.

All floor and table lamps, including reading lamps and bedside lamps, shall have their individual switches located in a visible area preferably on the base of a lamp or on the cable immediately after the base but not over the edge of the table. Visible cabling to be kept to a minimum. Switches on both sides of the headboard for bedside ambient light; reading lamps must be equipped with individual switches.

Provide 2-way master switch in entrance hall and on one side (Toilet Side) of the bed to control all lighting... All light switches on the master switch circuit to be fitted with relays so they can be individually overridden. It's mandatory for Radisson Collection, Radisson Blu and Radisson Red.

Or 2 Way switch (at Entrance and Bedside Toilet Side): One for ceiling architectural light and another two-way switch for indirect guestroom and standing lamp. The configuration will depend on the room design, brand and interior. To be defined as part of the mock-up room review and coordination.

Provide a switch to bathroom outside bathroom door, controlling all light fixtures. For larger rooms, an additional one needs to be evaluated.

Switching at the bedside should be simple and reduced to the minimum number on the headboard. It is recommended that no more than a triple plate is used on each side of the headboard containing a one gang master switch and a combined power outlet and USB-A&C 3.0 charger (see below).

Desk lamp preferably to have light switch one way at desk area. Also, for the bedside lamps to have on each side a one-way switch for the bedside light. Reading lights to be operable from the fixture.

08.06.2. Key Card / Motion Occupancy Sensor

Each guest room should have either a key card energy saver or occupancy sensor system.

Key card energy saver to be interfaced with all guestrooms lights circuits and power sockets except (TV, minibar, International Socket at desk and at headboard, and FCU Control). When the room is not occupied and the card is removed from the slot, all lights and other sockets will be off and FCU thermostat will go to eco or energy saving mode (pre-set temperature and off mode/low speed fan).

The same scenario and configuration will be applicable with occupancy sensors architecture where a sensor will be located at room corridor entrance and another sensor will be built-in the thermostat unit. The sensors need to be thermal to ensure the functionality when the guest is sleeping in the room.

Key card holder/occupancy sensor to have minimum 20 second time delay to permit guests to vacate the room, key card holder to have led light indicator.

08.06.3. Power Outlets

Power Outlets: Provide Ground Fault Interruption (GFI) as required by code and differential circuit breakers with a minimum protection of 30mA:

Finish and colors as agreed with the Interior designer.

Electrical power outlet locations must be coordinated with all furniture and equipment and provided as per the following:

- On the side of the bed closest to the window: -
 - Above the bedside table: -
 - 1 no. Normal Socket for guest use (24h).
 - 1 no. USB-A&C 3.0 outlet (24h).
 - Below the bedside table: -
 - 1 no. outlet for bedside lamp (switch on wire). Only if there is no pendant light.
 - 1 no. outlet for telephone if necessary.
- On the side of the bed toilet side: -
 - Above the bedside table: -
 - 1 no. Normal Socket for guest use (24h).
 - 1 no. USB-A&C 3.0 outlet (24h).
 - Below the bedside table: -
 - 1 no. outlet for bedside lamp (switch on wire).
- 1 no. universal outlet (24h) and 1 no. USB-A&C 3.0 outlet (24h) above the desktop.
- 1 no. outlet for floor lamp at the seating area (low level), controlled via light switch.
- 1 no. universal outlet (24h) for guest use at the seating area (low level) if required.
- 1 no. socket below desk for desk lamp.
- 1 No. twin outlet concealed behind the TV or TV panel (24h).
- 1 No. outlet concealed in the minibar cabinet for the minibar (24h).
- 2 No. outlet above the minibar cabinet (for kettle and coffee machine).
- 1 No. outlet in lobby area at low level for cleaners' use.
- 1 No. shaver socket in the bathroom.
- 1 No. IP rated outlet in the bathroom for heated towel racks (where installed in extreme climates).
- Additional power outlets as required by local code, equipment list and interiors design scheme.
- Additional power outlets as required in accessible rooms for various equipment together with additional sockets at bedside for medical equipment, etc.

Power outlet with USB: The USB-A&C 3.0 element needs to have its own overcurrent protection and must not rely on RCD or MCD devices to protect its circuit.

Guest outlets to be accessible without moving furniture with cover plates style/finish as specified by the interior designer.

08.06.4. Low Voltage/Data Outlets

Data outlets are to be provided as follows: -

- Wired internet connection above desk (for RB and RC only)
- One outlet below bedside table (typically on the side of the bed closest to the window) for the phone.
- One outlet at desk below desk level for telephone (Optional) (Depends on the classification and local requirements).
- One Outlet in bathroom for telephone (Optional) (Depends on the classification and local requirements).
- One outlet for automated mini bar fridge data connection (Optional).
- Two outlets for TV system connection concealed behind TV or TV panel.
- Additional as required in suites/larger rooms to suit layouts.

08.06.5. Lighting

Good lighting levels for guestrooms are extremely important and minimum levels as previously defined are to be achieved together with the right ambience within the room and the interior designer is to work with the lighting designer in achieving this effect.

Each guestroom has separate lighting and small power circuits, i.e., not duplicated with other adjacent guestrooms.

The appearance of the lighting equipment within the room is dependent upon the interior design but must also address issues related to durability, replacement, heat and safety.

Lighting shall be designed to maximize energy savings balanced with good visual effects. All lamps used within the guestroom must be LED.

In general, as a minimum the following lighting is to be provided: -

- Entrance lobby lights minimum 2no., surface mounted decorative fixture or recessed down light.
- 2 no. bedside lamps with separate reading light and individual switching. Lamp to ensure suitable lighting for reading and minimize the amount of reading light that might stray over onto the other side of the bed to avoid disturbing any accompanying guest.
- Desk area lighting to be suitable for reading/computer work, i.e., provide a desk lamp,
- Light over mini bar area, wardrobe facing into room.
- Pelmet lighting (optional).
- LED strip above/around headboard
- Floor lamp with individual dimmer switch.

In Bathrooms, lighting distribution should avoid strong contrast, shadows and poorly lit corners with minimum lux levels as previously defined with strong overhead lighting avoided. Minimum 3 downlights however final configuration may dictate additional spots; splash proof fitting with opaque lens within shower; built-in LED, warm tone, light fixture each side of mirror provided on separate switch to allow choice of mood/illumination level. Steam proof light fittings to be used throughout. All bathroom lighting to be dimmable.

Wardrobes to be well lit from the inside with LED lighting equipped with connected to a micro (limit) switch or motion sensor.

Scene setting dimmers to be provided in the suites.

08.06.6. Bathrooms

Switches are to be externally located subject to local authorities' requirements. Provide an IP-rated universal shaver socket outlet for guests to use adjacent the mirror.

Heated mirror demister pad to be provided linked to bathroom light switch together with backlit LED strip. In case of transformer to be placed with easy access and away from any water point. IP shouldn't be lower than IP44.

Magnifying mirror with built-in light with switch to be considered.

As an option a ceiling speaker can be installed as a slave unit to TV in bedroom with volume/off/on switch mounted on wall.

08.07. Lobby and Guest Public Areas

Allow sufficient power outlets for all decorative lighting and provide wall and/or floor outlets for guest usage in lounge seating areas, minimum 1no. 13A double outlet with double USB-A&C 3.0 per seating group of 5 with overall not less than 1no. double outlet per 50m² for guest usage. In the corridors, power outlets should be provided every 15m for housekeeping with additional outlets provided for ice machines, shoeshine, etc.

Provide a minimum of 5no. double power outlets at each reception desk with 3no. power outlets supplied via central UPS.

A simple scene-setting dimming system is required with a minimum of 5 pre-set lighting conditions to suit the lobby layout and light circuits distribution, to create mood light along different timings and settings.

All Public Areas with no lighting control (corridors, toilets,...) to be equipped with motion sensors.

08.08. Conference and meeting / support areas

Provide the following to each meeting room as a minimum and further additional in accordance with the interior design scheme, AV installation requirements and RHG IT guide requirements:

- 2no. 13A double electrical outlets with double USB-A&C 3.0 above all credenzas,
- For smaller meeting rooms with fixed style set up typically up to 40m² a minimum of 1no. 13A electrical outlets and USB-A&C 3.0 for each seat within the room with minimum of 6 data outlets and HDMI connection as per RHG IT guide. This to be provided through outlets mounted on or within meeting room tables,
- For all other rooms provide 2no. floor boxes each with 3no. 13A electrical outlets, 2 data outlets and HDMI connection (as per **RHG IT Standards**) in each room for the first 25 m² with 1no. additional floor box for every 25m² room area above 25m², minimum 6 outlets per room whichever is the greater.

- 1 double 13A electrical outlet with double USB-A&C 3.0 per 10m of wall to all room sizes,
- Provide electrical outlets above suspended ceiling at all windows for motorized blackout curtain,
- Power/Data outlets for AV equipment on projection walls and walls opposite the projection walls, for multipurpose rooms this would apply to all possible projection arrangements,
- 2no. data outlet adjacent to the projection screen and over credenza,
- 1 telephone outlet and handset in each room,
- Data and power outlet outside each room for digital signage,
- Data and power outlets for host/registration table location.

In addition to the above, the large conference rooms/ball rooms provide the following in each room:

- 2 x 16A 3 phase outlet.
- 2 x 32A 3 phase outlet.
- 1 x 64A 3 phase outlet (optional, based on M&E area requirements)

All floor box outlets must be covered with secure and robust metal lids with a locking mechanism to secure the lid both when the outlet is used and not used. Each floor outlet must contain a minimum of 3 power outlets and 2 data outlets.

The power supplies for AV systems must be clean, i.e., separate from the lights and other disturbances.

In addition to the above, electrical and data points within all pre-function and/or break out areas should be installed, with a minimum of 8m intervals along walls or adjacent to seating areas within floor boxes or within furniture elements for guest usage.

08.08.1. Lighting

Appropriate combination of LED decorative lighting with dimmer control/scene setting in each room and subdivision, suitable for business functions with minimum lighting levels as previously stated, all as per lighting consultants' design.

As there may be several lighting circuits, a simple scene-setting dimming system is required with a minimum of 5 pre-set lighting conditions to suit meetings, multimedia presentations, etc. together with interfaces to projection units. The system must be user-friendly, and easy for guests to use.

Provide a circuit with on/off switch for low voltage down lights in front of projection screen.

08.09. Kitchen / Laundry

The main workplaces in the kitchens (preparation, cooking, dishwashing, etc.) / Laundry (pressing, washing, Ironing, dry cleaning, etc.) must ideally have natural light and outside view wherever possible. Artificial lighting requires glare-free daylight fittings with minimum luminance of 500 lux.

Light boxes to be easily maintained, formed of metal cassette type light colored ceiling panels with integrated light boxes, LED, minimum clearance to floor 2800mm.

Lighting to be led by a warm tone that enhances food colors.

Provide panels outside of the main kitchen areas.

Provide ceiling outlets where needed e.g., island workstations etc. All the power outlets must be tamper-resistant and protected by RCDs.

08.10. Back-of-house areas (Offices, Stores, Technical Rooms, Pantries etc.)

Individual light switches should be allocated for each area with enough lighting distribution and fixtures. All lights to be LED and non-dimmable. 60x60cm LED panels, or LED spotlights can be considered for those areas. Provide occupancy sensors in BOH offices and storages.

08.11. External Power Outlets

Exterior electrical IP rated power outlets are required at the building perimeter at reasonable intervals, main entrances, restaurant/bar terraces and parking areas. In addition, ensure adequate provision of outlets for landscaping equipment.

08.12. Lightning Protection

Lightning protection to be provided to ensure safe protection of the occupants, the building and all its systems, complete with Anti-surge devices fitted at the Main LV switchboard.

The system shall consist of copper points mounted on cast copper bronze bases around the periphery of the roof. The points shall be interconnected by means of copper conductor cable to form a closed loop. From the closed loop, the down conductors shall be coursed down from each level. These conductors shall be bonded together where they enter and leave the core area. From the base of the core area, the conductors shall be run to the periphery of the main structure and coursed

down the exterior columns behind the façade and terminate in ground rods outside the foundation walls. One of the grounds shall be interconnected to the switchboard room ground bus.

In dry soil areas, a counterpoise conductor connecting all ground rods is recommended.

08.13. Aircraft obstruction light

Where local regulations require or it is required by RHG D&T the system shall be installed consisting of two lights mounted at the highest point on the roof, arranged in twin lamp combination, complete with mounting post and waterproof distribution box. The warning lights shall be controlled by a time switch to put the lamps on one by one automatically. The system shall have a current relay mounted adjacent to the lamps to indicate current flow when lamps are on and sound on alarm if there is no current flow when switch is in the "On" position. When an alarm occurs, the systems should be capable of switching over by means of auxiliary relays from one lamp to another. System to fully complete with all applicable regulations.

08.14. Server/Data room

Provide total protection by earthing and bonding of the IT equipment according to ANSI/J-STD-607, IEE1100-2005.

09. FIRE LIFE AND SAFETY

09.01. General

Radisson Hotel Group Life Safety Standards have been established with the aim of providing a safe environment for the occupants of any Radisson Hotels.

Fire Life safety system must include the requirements described in the following sections of this document. Where local governmental codes require alternative or more stringent requirements, these differing requirements must be complied with by the designer. Where this situation exists, Radisson should be notified in writing of any change in the system being specified from this standard, the reasons for the change and a full description of the variation prior to any implementation of the changes.

This document is not a construction specification. It is an aid to designers to achieve a uniform and consistent approach to life safety in RHG properties throughout the countries where Radisson brands hotels are present.

This document should be read in conjunction with the approved Fire Strategy that has been developed by a suitably certified Fire Engineer, depending on the local regulation.

Earthing – bonding: In the mechanical/technical rooms care shall be taken to evaluate the need of earth bonding in the AHUs, cable management systems (cable basket trays, racks, etc.), incoming water steel pipes etc.

09.02. Building Requirements

09.02.1. Security Measures

Site planning is of key importance for the protection of life, property and operations and a security consultant together with the RHG Safety and Security department must be consulted to give all necessary advice and any approvals. The security plan must consider all vehicles, pedestrian, team members and guest circulation, infrastructure, landscaping, controlled zones and safe distances, physical protective barriers, parking, loading and unloading security, locational risk factors, etc.

Additional means might be required, subject to specific local requirements such as earthquake or hurricane risks.

It is strongly recommended for all locations and it is mandatory for those locations considered to be high risk, that the requirements as issued by the Federal Emergency Management Agency (FEMA) or local governing body, are reviewed and implemented as applicable following a detailed assessment of the risks present. RHG will confirm the detailed scope for the 3rd party assessment.

09.02.2. Compartmentation

The layout of the hotel shall provide acceptable compartmentation through use of fire doors and fire screens at strategic locations to restrict the spread of fire/smoke. Fire resistance for the structure and floor to floor compartmentation to be a minimum of 30 minutes and for buildings above 3 levels this requirement is to be increased to a minimum of 60 minutes. For other areas the following fire resistance separation to be provided as a minimum standard, however Local building control legislation may dictate greater fire ratings.

- Between the upper floors and basement 60 minutes,
- If the parking area underneath the hotel, it's needed to increase to above 120 minutes with access staircase from the parking separated by a set of 2 separate doors with minimum E60 fire resistance,
- Technical shafts to be fire separated from evacuation routes with minimum 60 minutes,
- Elevator shafts to be a minimum 60-minutes fire rated and separated from evacuation routes by means of fire doors.

Compartments to be planned as per local regulations but irrespective escape routes maximum travel distance before reaching a secure area with minimum one vertical escape per compartment for a normal fire risk situation to be as below chart. For high-risk situations this may be reduced by 50% on guestroom floors.

Travel Distance Requirements				
	without Sprinklers		with Sprinklers	
	Preferred	Maximum	Preferred	Maximum
Travel distance from a guestroom door to an exit	40m	46m	55m	61m
Travel distance to an exit for all other areas	50m	61m	70m	76m
Maximum dead-end corridors	10m	15m	15m	20m

All escape routes must be visible and easily accessible from any position within the hotel. In planning the compartmentation of the hotel every guestroom must be designed as a safe area with 60 min fire rating and main guestroom door with 30min fire rating.

In planning the escape routes, the following are to be considered: -

- confirm that the number and location of existing exits is adequate by reviewing the no. of people on each level and the distance they need to travel to reach the exit,
- Travel distance is measured from all parts of the premises to the nearest place of reasonable safety which is to a:
 - protected stairway enclosure/story exit
 - separate fire compartment from which there is a final exit to a place of total safety, or
 - nearest available final exit.

For exceptional reasons the minimum travel distances may be increased as agreed with RHG D&T by the addition of further fire protection measures, e.g., automatic fire detection, sprinklers, etc.

Corridors to be compartmented by fire rated compartment doors or smoke doors. Each section with a staircase shall be isolated and possibly additional doors will be required, i.e., dead ends, etc. Fire detection systems shall automatically close smoke doors.

Compartmentation might not be permitted in certain instances, such as corridors where mechanical ventilation is required, and in this case other suitable measures should be considered and agreed together with RHG D&T.

Fire dampers are to be installed as required in accordance with general good practice. Fire dampers shall also be installed for all ducts/conduits exceeding 125mm diameter where they cross through any compartment. All such dampers are to be shown on the as-built drawings. All fire dampers must be provided with an external device which clearly indicates the position of the damper with easy access for resetting and maintenance.

All dampers should be reset automatically unless easily accessible in which case manual reset can be used. Fire damper position indication to be indicated on the BMS. Self-motioning as part of the test/maintenance procedures to be scheduled through the BMS on a weekly basis. Fire damper motors to be in line with the fire rated class.

The kitchen exhaust system must be an individual system serving only kitchens on one floor and must not be interconnected with any other building exhaust system, fire suppression system and grease separators.

The range hood exhaust air volumes must be sufficient to provide for capture and removal of grease laden cooking vapors.

09.02.3. Riser Shafts

Riser shafts to be enclosed at basement and attic levels to avoid vertical spread of smoke. For high rise buildings of 6 floors or more vertical separation shall be on every floor including basements including all fire collars, etc. The shaft itself must be fire resistant and sound proofing with penetration of pipes must be carefully sealed/fire stopped. Easy access to technical shaft which is to be secured by a by self-closing 1-hour fire resistant lockable door complete with an approved "bit lock" key. Electricity and data wiring are not allowed in shafts containing pressurized water pipes and are not recommended in shafts with drainpipes.

09.02.4. Dead-Ends

Dead-end situations to be avoided. However, if no alternative solution exists, then dead ends up to a maximum 15m without sprinkler system (mentioned above in travel distance chart) on guest rooms floors may be permitted (measured from last guest room door), with additional safety measures required such as manual call points, hose reels, etc.

09.02.5. Staircases

Generally, stairways must be at least 105cm wide and, in any case, not less than the width of the escape routes that lead to them. Stairways wider than 2100mm to be divided into sections, each separated from the adjacent section by a handrail, so that each section measured between the handrails is not less than 1050mm wide.

An alternative means of escape is required from all areas of the hotel with a minimum of 2 escape routes per floor via protected/enclosed escape staircases that allow direct egress to the exterior or exit via a protected area on the exit floor with maximum travel distance to an escape staircase as stated above.

External staircases are permitted subject to the height of the building. However, additional protection may be required and should be safe distance from any openings in the facade to ensure provides a safe escape route.

Ensure adequate emergency signage and escape route lights are provided, and each landing has the floor level clearly indicated.

On the exit level additional measures are required to avoid external access into the staircase but still allowing non-obstructed egress, i.e., panic hardware, monitored/alarmed, etc. In situations where staircases are shared with other building occupants it must be ensured that access to the hotel floors from the staircase is restricted only allowing escape in the event of an alarm being activated and must be both alarmed and monitored for security.

The escape routes outside the building should always lead to an agreed safely navigable assembly point. Assembly points to be located away from the hotel at minimum 1,5 times the building height in order to avoid falling debris or structural collapse, or any safe place agreed and confirmed by the emergency team of the hotel.

09.02.6. Secondary Escape Route

All PUAs and BOH areas of exceeding 50m² or having occupancy of more than 19 persons are to be provided with 2 separately distinguishable means of escape. If these escapes open into the same corridor, compartmenting doors must separate the escape routes subject to RHG D&T approval.

Primary and secondary escape routes are to be treated technically as the same in terms of dimensions, fire rating and other requirement of pressurization, smoke management, emergency light, etc.

09.02.7. Lifts

Guest lifts to be located in area clearly visible from reception. Car park lifts shall be separate lifts with travel only to the reception floor. Lifts accessing general PUAs such as meeting/function areas must be separate to the main guest lifts. All lifts have key card access control.

All lifts to be also interfaced with Fire Alarm in case of emergency and evacuation to be landed to ground floor or an alternate floor as per the fire location. Only fireman lifts can be used by the specialist (Fire brigade, Civil Defense, Emergency Team) during evacuation and emergency events. Fireman lift to be equipped with fireman switch, shaft pressurization, emergency lighting, generator backup connection or alternate incoming electrical mains supply with intercom system.

09.02.8. Accessibility

Special provision for guests with disabilities as per local regulation. At minimum, a stroboscope light shall be fitted in the accessible guestrooms, a vibrating pillow, with pull cord alarm at bedside and in bathroom to be also considered, and LED indicator at the main door at the corridor. Accessible toilet in PUAs to be also equipped with pull cord alarm connected to reception.

A fire evacuation wheelchair is to be provided in the floor staircase where accessible rooms are located to be used in case of evacuation by the evacuation team.

09.03. Life Safety Systems

09.03.1 Fire Alarm System

Fire detection and alarm system is to be state of the art, fully automatic/addressable, designed according to the latest approved fire safety innovations with features fully compliant with current fire safety regulations, all in accordance with NFPA or CEN standards. Fully automatic in that the system does not require human interference to raise the alarm and evacuate the hotel guests and team members. System may have functions that allow trained personnel to interfere with the alarm sequence and function but in the absence of any human reaction or interference the system must still automatically alarm and evacuate the hotel guests and team members. Fully addressable in that any fire detector can be identified by an individual address and room description, and any activation of fire detector or manual push button will show on the fire alarm panel identifying the room (room number), floor and detector number.

Fire alarm panel for guest rooms, PUAs and BOH with information and identification display and printer is to be provided with the Back Office. If a security office is provided within the hotel, then the Main Fire Alarm panel is to be located here with a repeater panel located in the Back Office/adjacent reception with signal audible from the reception desks, and a graphic display PC to be located in the engineering control room. 2-sets of A3 size laminated fire alarm zone and detector plans are to be provided easily available for fire fighters within reception/security office. The main security control room will be the fire command center during evacuation to manage and control the evacuation process for the safety of all hotel occupants.

The fire alarm must, wherever possible, be directly connected to the local fire department or 24-hour emergency monitoring service. The connection shall allow for an adjustable time delay of up to 5 minutes before automatically evacuating or alarming the guests (the delay depends on the configuration of the property and must be agreed in cooperation with, and approved by, the local authorities and with RHG D&T). However, there must never be any delay alarming the appropriate team members and a push button override must be provided at the main panel and repeaters to allow for immediate evacuation.

Fire alarm system shall automatically close smoke and fire doors, close down ventilation plants according to the building fire strategy and agreement with the fire authorities and should be interfaced to:

- BMS for monitoring of AHU's, smoke exhaust systems, pressurization fans, kitchen exhaust and fresh air, laundry exhaust and fresh air etc.,
- Telephone for remote alarms and WIFI / DECT phones interfaces,

- Paging via WIFI / DECT phones,
- Access control systems to allow easy evacuation and entrance for the fire brigade,
- PA system for automatic announcements,
- Background Music, TV system to switch off in an emergency, and AV systems for meeting rooms,
- Gas Detection and Gas solenoid,
- Fire suppression systems (FM200, CO₂, Clean Agent, Ansul etc....)
- Elevators (Passengers and Service), escalators, etc.,
- Kitchen power sockets (Main kitchen equipment) incl. automatic switch-off in event of fire alarm
- ZCV (Zone Control Valves) in all floors and branches.

Before handing over to RHG the system must be fully tested and certified by an internationally accredited independent specialist.

All Interfaces of MEP systems and equipment with fire alarm must be via interface modules and not through BMS as per the code.

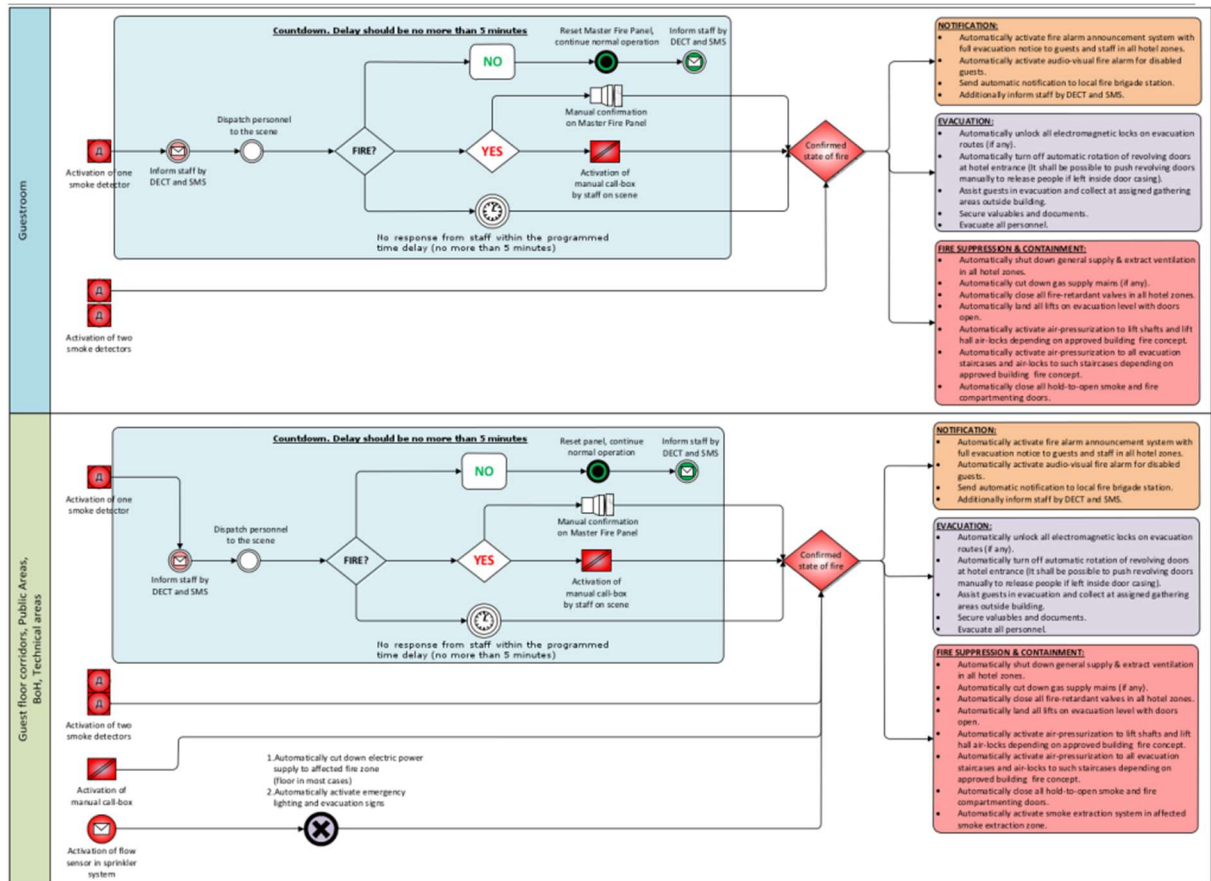
09.03.2 Smoke/Heat Detection

For smoke detectors only, the connection shall allow for an adjustable time delay of up to 5 minutes before automatically evacuating or alarming the guests. The delay needs to be defined in cooperation with, and approved by, the local authorities.

There must be no delay alarming the reception team members. The possibility to override the time delay by push button must be provided to allow for immediate evacuation button.

The system shall include but is not limited to: -

- A minimum 1 smoke detector is required per guestroom if others are not requested by local legislation and if the guestroom has more than one room it should be more than 1 detector / with built-in sounders. Detectors are not required in bathrooms,
- Smoke detectors in guestroom corridors, stairwells, raised floors, etc.,
- Full coverage with smoke detectors in PUAs including parking and heat detectors where appropriate (saunas, steam rooms, etc.),
- Full coverage BOH with smoke detectors and heat detectors where fuel burning equipment or appliances present (i.e., kitchens, plant rooms, etc.),
- Smoke detectors and two-step heat detectors in computer and PABX server rooms,
- Full coverage in suspended ceilings, lift shafts, vertical ducts, etc. with duct detectors (>3400m³/h) incorporating remote switches and indicator lights.
- Heat Detectors to be located in Kitchens, Laundry, and Boiler rooms etc.,
- Beam Detectors in atrium and high ceiling floors.
- The following algorithm is recommended to implement, however local legislation prevails.



09.03.3 Gas Detection

Gas detectors as appropriate are to be fitted in all locations where equipment that utilizes any gas, or where there is the possibility of gas leaks occurring and on the routes of the supply ducts together with an automatic cut-off systems installed at appropriate locations. Detectors to be fitted at suitable height subject to the type of gas present. Ensure easily accessible, clearly marked shut off valves are fitted at access to the building and near the respective plant rooms, kitchen, etc. Gas system to be also interfaced with Fire alarm system to shut off complete gas supply to the building in case of evacuation.

09.03.4 Carbon Mono and Dioxide Detection (CO/CO₂)

Areas with the presence of equipment and vehicles with CO and CO₂ emissions must be properly controlled with a detection system.

In order to control ventilation to maintain a healthy and acceptable level of carbon within the areas. Areas mandatory to have the system are underground parking, large event rooms, and theatres. Also demand ventilation for building floors and PUAs can be of a sustainable option to save energy and maintain healthy level of carbons within the building.

09.03.5 Manual Call Points

Manual call points to be provided to local regulations, but at minimum to be fitted adjacent to each exit from a compartment. Additional points might be required as deemed necessary such as reception, dead end situations, etc. As a minimum emergency call button should be provided in the sauna, steam room, fitness, pool area, accessible rooms.

09.03.6 Notification Alarm/Public Announcement System

Emergency notification alarms must be provided in addition to the emergency alarm that is required by local legislation. Ensure notification is audible throughout the property including guestrooms with emergency announcement override provided to all areas including guestrooms. The emergency announcement shall be activated automatically by the fire alarm system and manually by reception employees with 4 pre-recorded messages (Alarm, False alarm, Test is planning, Test is ended) in all local languages plus English, with others as required depending on location.

A zoned distribution is to be provided for all areas with zoning of the emergency announcements done in cooperation with the local fire authorities, the alarm system suppliers and RHG D&T. Announcement zones are to be as per the evacuation zones to enable phased evacuation as permitted by local regulations.

Loudspeakers must be placed so that information is clearly audible in all areas of the hotel (minimum 60dB), with a separate announcement loudspeaker located in each guestroom or can be replaced with a built-in sounder base in the detector. The announcement volume shall be 70 dB with articulate voice distinction at the bed pillow for all guestrooms in addition to a strobe light in all accessible rooms. A command microphone with zoning possibilities shall be located near the fire alarm panel and at reception for use by hotel team members.

09.03.7 Sprinklers

RHG strongly recommends that sprinklers be installed in all properties. However, this is mandatory in high rise buildings higher than 23m (ground level to floor level of highest floor) or more than 6 floors, and/or airports hotels.

Sprinkler systems must be designed according to sprinkler codes such as NFPA 13, CEA 4001, EN 12845 or FM. High-pressure water mist/fog fire protection system is to be considered and is generally the preferred option, where allowed, in terms of the local codes.

For low rise and medium rise buildings installation of sprinkler systems will depend on code, structure, fire compartments, actual height of building, layout, etc. and shall be determined on a case per case assessment. In certain markets irrespective of any statutory requirements, sprinklers may be required depending on local competition, particular market factors, etc. Final decision regarding sprinklers to be discussed and agreed with RHG.

All parking areas are to be fully sprinkled. The sprinkler pump system consists of primary pump sets together with a jockey pump plus full backup pump sets that switch on automatically in case of primary set failure. All pumps will be required to be tested on a regular basis and test lines shall be provided. Test lines to be located within plant rooms.

Fast response heads, dry pipe sprinkler system in non-heated areas only with recessed sprinkler heads in PUAs. Ensure fire resistant material is used for piping. Valves, flow detection, test valves to be fitted. Ensure loops per floor are designed.

In designing the sprinkler system, the following requirements apply: -

- Perform water flow test/s and record results to ensure reliable source of water volume and pressure from local authority water main or from onsite cisterns/tanks if local supply is not available or unreliable. Provide sufficient water tanks or water reservoir for occupancy and maintenance,
- Allow a 10% hydraulic safety factor,
- Design system within maximum pressure of 12 bar without use of pressure reducing valves,
- For low rise buildings provide control valves, flow and tamper switches at each connection monitored on the fire alarm system. Locate inspector's test assembly at remote end of zone (not in guestrooms) with discharge piped to drainage system,
- For high rise buildings connect guestroom floor sprinkler piping to 2 risers at each floor with a control valve and flow and tamper switches at each connection. Locate inspector test assembly as above,
- Sprinkler heads generally to be fast response 68° to 74°C except as for certain locations as agreed with RHG D&T. Sprinklers to be either recessed, semi-recessed and concealed throughout,
- Provide wet-pipe sprinklers for habitable spaces such as guestrooms, guestroom corridors, restaurants/bars, ballrooms/meeting rooms, public and back-of house areas,
- Provide dry-pipe sprinklers in attic and unheated areas to avoid the possibility of freezing,
- For guestroom bathrooms less than 5.5m² sprinklers are not generally required and where non-combustible materials are used,
- Provide sprinkler protection to all underground parking areas, and plant rooms,
- For loading docks/truck bays provide sprinkler protection, with dry-pipe system if area is subject to freezing,
- For freezer/cold rooms protect with dry-type heads supplied off the area wet sprinkler system.
- It is recommended to use flexible sprinkler fittings intended to connect the branch line to the sprinkler head by utilizing a flexible stainless steel hose assembly and steel bracket assembly.

Sprinklers are not required in wardrobes, dressing rooms or stores where the area of the space does not exceed 2.5m² and does not contain a washer, dryer, water heater, or any other mechanical/electrical equipment.

Sprinkler installation is allowed in electrical rooms where equipment is 600V or less, except for the Dedicated Electrical Space.

09.03.8 Smoke Control

Subject to local fire authority all staircases to be fitted with natural ventilation to top of the stairwells, opening automatically either when manually activated or when fire alarm is activated, or staircase smoke detection is activated.

High rise buildings above 6 floors to have pressurized ventilation system and lobbied staircase access together with smoke extract system in all fire escape routes and PUAs.

Ventilation might be required in atria or similar type structures/areas as recommended by the appointed fire consultant.

Ensure basement floor parking is sufficiently ventilated to avoid smoke build up in the event of a fire.

09.03.9 Fire Fighting Equipment

Firefighting equipment is to be clearly marked and easily accessible, preferably near the exits from each compartment and must be installed in cabinets, locked with break-glass devices. Firefighting landing valves to be positioned throughout the hotel as required under the fire code.

Fire extinguishers to be provided as required by local regulations/fire authorities with appropriate types for intended use, i.e., CO₂ near electrical rooms, powdered avoided near IT equipment, etc.

Portable 50kg fire extinguishers are to be provided within the car park securely mounted on the walls.

Fire stations fitted with fire hose reels and extinguishers to be provided with adjacent fire exit on all floors, minimum 2no. at not more than 30m apart.

- IT rooms must not have sprinklers installed, a clean agent or Fm200m can be accepted,
- Diesel Storage to have Foam suppression system, Gas Tank deluge system to be considered,
- Electrical rooms to have suppression CO₂ system.

09.03.10 Kitchen Fire Suppression/Exhaust Hood

Fire blankets to be fitted in kitchen areas.

Provide fire protection in accordance with local statutory requirements code but as minimum designed according to NFPA 96 (Standard for Ventilation Control and Fire Protection of Commercial Cooking Operations), ANSUL R102 or similar. Fire suppression systems to be built into the exhaust hoods over cooking equipment connected to fire alarm system with spray nozzles to be provided within all ventilation structures, exhaust hoods, plenum and ducts, filter surface protection and over cooking equipment such as deep fat fryers, griddles, grills, natural charcoal or chain type broilers, electric lava rock grill and all gas equipment, etc. Discharge the fire suppression agent to be activated by means of fusible links, as well as by manual action. In the case of fire alarm, the central gas solenoid valve and/or the central electrical shut-off power device will shut off all supplies to the area affected together with emergency shut off at exit to kitchen.

The central gas solenoid should be connected to BMS through fire panel for communication.

Ensure good access provided for cleaning of ducts, grease traps, filters, etc. to avoid build-up of grease.

09.03.11 Fire Hydrants, Fire Hoses & Extinguishers

Fire hydrants, fire hoses and extinguishers shall be installed and located as per local regulations but irrespective hydrants and extinguisher stations to be clearly indicated and located minimum 30 m apart. In addition, dry risers to be provided where regulations allow together with suitable couplings for connection of local firefighting equipment. Breeching Inlet to be located at ground floor with truck parking space and to be connected directly to the firefighting risers.

Whenever fire hydrants, fire hoses and extinguishers are exposed to potential damage from vehicles, additional protection is required.

09.03.12 Door Closers

All compartmenting doors and fire doors are to be fitted with door closers/automatic release devices, including all guest room doors, staircase doors, housekeeping areas, BOH areas with access to PUAs, main plant rooms, etc.

Double doors which are fire doors must close in the correct sequence and doors must provide an effective smoke sealing.

09.03.13 Panic Hardware

All final escape doors, i.e., doors leading to the exterior, to be fitted with panic hardware. Panic bars are mandatory in areas where guests are likely to evacuate during the night, such as staircase exits and in meeting rooms larger than 300m².

10. SECURITY

10.01. General

Access control is to be considered on an individual basis subject to location, configuration, local risk assessment, etc. with all access to the hotel well controlled however with due consideration to maintaining a positive guest arrival experience. Main components of access control systems to comprise key card access, CCTV monitoring, alarm intruder detection linked to CCTV, scanners, etc. employing latest best technologies.

10.02 Monitored Access Control

Provide an access control system for the personnel entrance and other sensitive areas of the hotel, e.g., computer room, storerooms, certain offices, server room, luggage room, security room, gate barrier, elevators etc. All access doors from PUAs that lead into the guestrooms, e.g., fire escape stairs, to have key card access. Access control systems must only be used in accordance with local laws and regulations and must be able to identify and register all persons passing both in and out of controlled doors. Systems to include intruder detection and forced entry alarm.

10.03 Exit Control / Security Checks

In certain locations, a dedicated Security control office may be required as confirmed by RHG. Where this is not a requirement, then the system shall provide monitoring as below to designated manned area as agreed with RHG D&T.

Every exit door of the property, including fire exit doors, fits with electrical alarm contacts to indicate open door position and linked to the CCTV system. Visual and audio alarms for these doors to be monitored in the security office or other designated manned area as agreed with RHG D&T. Remaining fire staircase doors in floor sot be monitored via door switch and to be connected to MIMIC panel at main security control room.

At main entrance points, it may be necessary to provide airport security style scanners/detectors for persons/luggage built into the entrances so as not to be obtrusive to the guests. In certain locations a security room for spot checks to be located within the reception area. Irrespective provision is always to be made within the design of these areas to allow for possible retrofitting of such security measures.

An electronic guard tour recording system with check points to ensure guards are active and accountable during patrol rounds duties 24/7 are to be installed.

10.04 Lifts

Key card access control to all passenger lifts is mandatory.

10.05 Restaurant Access Control

Depending on location, the restaurant door may be required to be fitted with an electrically operated magnetic lock controlled from a panel behind the bar with alarm function at reception desk.

10.06 Car Park Access Control

This area to be part of the site planning, with controlled access, remote check points and spot check points on the entry roads in and out of the hotel.

Assurance of all access to all parking areas security controlled is needed, by the installation of two automatic lifting barriers for entry to and exit from the internal road system. For underground parking provide automatic heavy traffic type door with hydraulic controls and baked enamel paint finish.

Provide guest key card and/or paper ticket functioning control for the operation of the overhead door/barriers to all car parks. In addition, also provide access control to parking areas with key card readers on connecting doors with additional two-way phones/intercom to reception.

Provide parking payment machine in the Lobby interfaced to PMS. CCTV to provide full coverage of all car parking entrances and parking areas.

In higher risk locations retractable bollards and protection screens must be installed, as per the site layout plan.

10.07 Fire Brigade Access Control

Access control systems to allow easy evacuation and easy entrance for the fire brigade, i.e., entrance barriers, etc. A complete set of emergency files containing as built drawings, electrical installations, HVAC, FLS systems, etc. is to be prepared and located in a dedicated panel adjacent the main fire panel for use by the fire authorities.

10.02.CCTV - Video Surveillance

10.02.1. General

CCTV system installed in accordance with local requirements/legislation with due consideration to any privacy issues. CCTV must cover all high-risk areas in the premises as stated below. For certain locations and markets CCTV coverage may be required in all guestroom corridors. Most recent **RHG CCTV Standards** must be taken into consideration for design purposes.

CCTV must be a closed-circuit digital system with high-definition color cameras throughout. CCTV monitors should be provided in the back-office area or within the dedicated security office which is typically provided in larger hotels/resort locations. Storage units, video servers and other sensitive equipment must be placed in a well-protected, access-controlled area such as the IT-server room. Cameras and lenses must be chosen with regards to lighting, range etc. to ensure that they provide high level quality images to ensure persons/objects can be readily identified.

Surveillance camera input shall be recorded with digital recording that allows full management of captured images. In addition, the following requirements are to be adhered to:

- Inform guests and visitors about video surveillance, through proper signage.
- Check with the relevant authorities if the local legislation requires a license, permit or other authorization and approval for the use of video surveillance.
- Include information about video surveillance in contracts or guest registration cards in order to obtain a "prior approval" from guests and clients.
- A standard sentence informing about video surveillance should be included in all employment contracts.
- Make sure your owner is also aware of and approves the use of surveillance.

10.02.2. CCTV Locations

Mandatory locations:

- Parking areas
- Loading areas
- Main entrance
- Service entries
- Team members entrance
- Reception facing guests (hold-up coverage)
- Reception facing desks (for check-in theft)
- Ground floor fire escape stairwells (to cover unauthorized use of emergency exits)
- PUAs as in lobby, bar, banquet foyer/breakout areas
- All leisure areas, including fitness rooms, swimming pools, but excluding areas where privacy required, i.e., changing facilities, sauna/wet areas, treatment rooms, etc., all subject to local regulation
- Basement and parking area elevator (lift) lobbies
- Lift lobbies and lifts
- All cash handling and POS (Point of Sales) stations
- Inside IT server rooms (to ID persons in/out and ideally facing servers)
- Inside baggage storage facilities (facing luggage)
- Drop safes & safe deposit boxes
- Fire exits
- Pre-function areas
- Any other key areas as specified on a project per project base
- For certain locations and markets CCTV coverage may be required in all guestroom corridors.

Optional Locations:

- External facade, grounds, perimeter and driveway access
- BOH areas such as important storage facilities
- Main circulation nodes

10.02.3. Installation

- Storage units, video servers and other sensitive equipment should be placed in a well-protected, access-controlled area such as the IT-server room or within the dedicated security control room (typically available in larger hotels/resort locations/high risk locations)
- Cameras and lenses should be chosen with regards to lightening, range etc. in order to ensure that they provide maximum benefit if identifying the images that they capture.
- Surveillance camera input shall be recorded with digital recording that allows full management of captured images.

10.02.4. CCTV Specifications

a. CCTV Monitors & Recordings

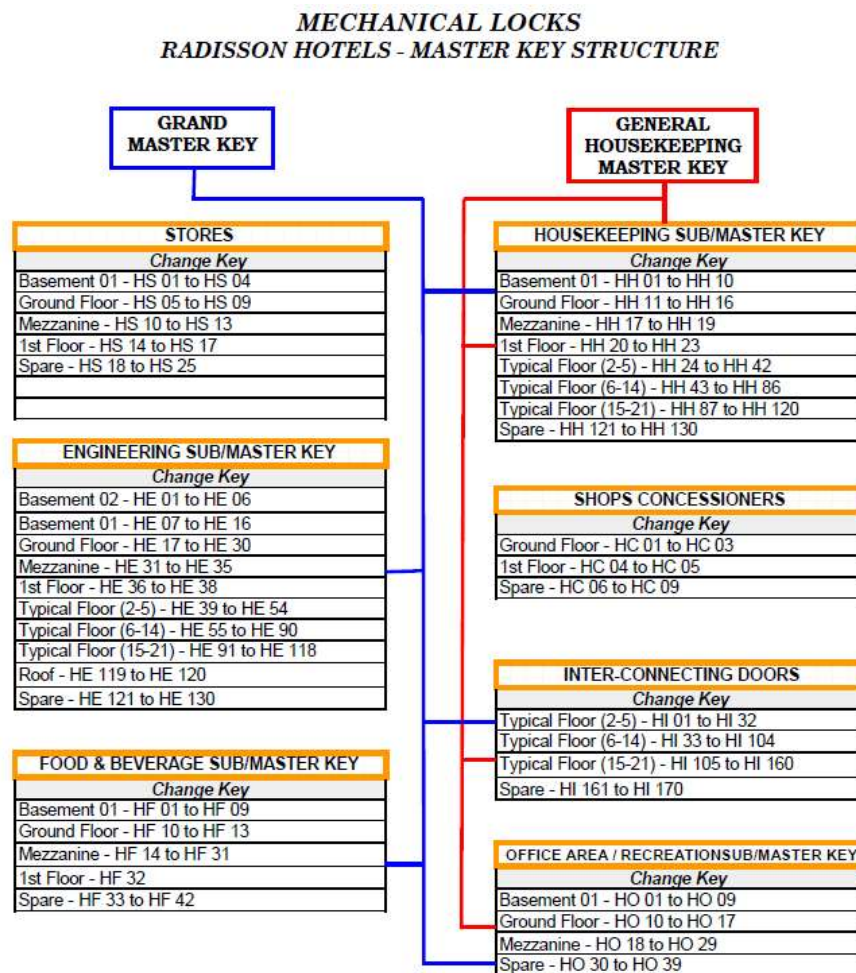
- The system shall be able to record for 31 days **without human assistance**
- Images to be stored **for a minimum of 1 month**
- IT room where the record must be kept **for 3 months** for PCI compliance,
- or as long a period as is legally acceptable
- Spot monitor at reception desk live view on external access points evening/night team

b. CCTV Minimum Hardware requirements

- 4,2 Mega pixels
- HD Color
- Dome, activated on movement
- Placement to be able to see height, faces.
- Alarm intruder linked to CCTV: need to detect open door positions

10.03. Ironmongery

Ironmongery and Door Locking: Mechanical Master Key System Architecture and Policy should follow the following diagram:



10.03.1. Guest Room Locks

Guestroom doors should have electronic card locks all as per **RHG IT Standards** with the following but not limited to features:

- Minimum 20 channels/virtual locks with interrelation possibilities.
- Selection of lock mode between normal, passage and internal.

- Real time operation.
- Lock interrogation, minimum last 100 entries.
- Battery powered with 3-year lifetime and fitted with low battery warning to team members only.
- Single action panic release cylinder with deadbolts. Automatic deadbolt is preferred.
- Lock system CPU to be interfaced to PMS.
- The electronic card lock/system must have single and multi-track encoding for possible future upgrades.

Special conditions apply where locks are used on exterior doors in extreme conditions, i.e., heat, dust, cold, humidity, etc.

The night latch mechanism is recessed into the door, this reduces the fire rating of the door. In order to ensure the correct fire rating is maintained the mechanism must have an intumescent wrap fitted.

In case of a potential installation of a Digital Door lock system, it needs to be validated by **RHG IT Department**.

10.03.2. Front of House Locks

The locks as above to be used on all doors where guests/visitors have access during their stay including all meeting room doors, health club, parking, etc.

10.03.3. Back of House Locks

All doors should have locks as described above. However, manual locking systems are only allowed in utility rooms where access is required by local authorities.

10.03.4. Background music system

Main BGM Rack located in the Server room and interfaced with Music Stylish to supply music to the following areas:

- Lobby / Reception with separate Volume Control localized at reception desk for lobby speakers.
- Drop Off outdoor area to have speakers for music for guests while waiting for car delivery.
- All Day dining restaurant with separate volume control located in the ADD area.
- The Business Lounge with localized volume control located in the area.
- BAR Lounge with Volume Control and Channel Selector, located in the area.
- Gym to have separate music rack with amplifier and CD Combo localized in the gym area.
- Massage rooms /Treatment to have provision for iPod Dock.
- Ballroom Pre-function area.
- Canteen with localized volume control in the canteen area.
- Board room with localized volume control in the area.
- Pool Terrace and Lounge area on roof to have volume control and channel selector localized in lounge area.

Please see **RHG IT Standards** for extra details.

11. DOORS

11.01. Entrance Doors

Main Entrance to be a large revolving door minimum 330cm diameter of three panes each minimum 160cm, or a pair of automatic sliding doors minimum 3.6-3.8m wide, distance between set of doors not less than 300cm, all controllable from the reception.

With sliding doors, special attention to be taken into consideration in the design, for ensuring the functionality with unfavorable weather conditions and air drafts. The sensors of both doors should be linked, and have reduced scope, to minimize the overlapping of the opening of the doors. Timing adjustment and weather mode with different functionalities should be installed. The aim is both Energy reduction, and guest comfort. With extreme weather conditions, the system would allow the possibility of keeping closed the inner door, until the outer door is closed, and vice versa.

Doors should be frameless, fully glazed aluminum/steel doors with security glass, and aluminum insulated hollow door frame with proper seals, insulation, etc.

11.01.1. Hot/Cold Air Curtain

Recirculating or non-recirculating air curtains should be installed in the main entrances in order to create a barrier inhibiting the heat transfer from one zone to another. The air stream shall be preferably not heated/cooled. The door opening must be completely covered by the stream of air generated by the air curtain. The airflow has to be controlled and adapted to the outdoor climate.

Hot and/or cold air curtains, depending on climatic conditions required, in order to protect against draft from all entrance doors.

11.01.2. Swing Doors Next to Entrance

Swing doors beside main entrance doors are required for luggage handling and wheelchair user access, to open outwards in the event of an emergency situation.

11.02. Kitchen Doors

Kitchen doors shall be solid core, constructed of non-corrosive material, lockable, 2-way swing, suitable for roll-through traffic with kitchen trolleys with stainless steel kick plates and view panel. In some locations, like entrance/exit from the kitchen, they must be separated.

Depending on fire regulations, doors may be self-closing or held open on magnetic contacts. Doors shall have no threshold, but heavy-duty rubber-sweeping gasket.

Main service doors into restaurant to be opened by kick plate, infrared sensor or foot switch on both sides in/out. Care to be taken to provide attractive finish on the PUAs side, matching the interior design scheme and recess the doors to prevent views into service areas.

12. LIFTS

12.01.General

Vertical transportation and lift selection with specifications should be supported with the following reports and documentation when the project outline has been agreed:

- Traffic Study
- Detailed Performance Specification
- Detailed Tender and BOQ document

A qualified elevator engineer working for the project's appointed Consulting Engineers should be used to execute the above.

The number, size and speed of the guest passenger elevators shall be based on a calculated average interval during peak periods, of 30 seconds or less (waiting time), and a handling capacity of 15% of the Hotel occupancy population in a five-minute period. For this purpose, "Hotel occupancy population" shall be assumed to be 1.5 times the number of rentable guest units in the Hotel based on 100% occupancy.

The elevator traffic study shall be prepared by a qualified and experienced elevator engineer, and the assumptions used in the calculations shall be approved by the RHG technical team.

The number, size and speed of the room service elevators will depend not only on the size and configuration of the Hotel, but also on the type and quantity of room service expected, and the locations of the service centers using the elevators (room service, housekeeping department, maintenance shops, etc.)

In general, there shall be no less than two (2) service elevators in hotels of up to 200 rooms, no less than three (3) in hotels of 201 to 600 rooms, etc. (The number of service elevators should be approximately 50% of the number passenger elevators.

Special service and special freight elevators shall be provided, as found necessary in planning the Hotel, to expedite the movement of goods and/or maintain security.

In all cases, each installation must consist of at least a pair of elevators, to ensure continuity of service during periods of maintenance and repair.

The elevator installation shall comply will all requirements of local regulations and Codes, and with the latest edition of the American National Standard Safety Code for Elevators (ANSI) A17.1 or British / European Standards, where these provisions are more than the local Codes.

12.02.Equipment Summary

12.02.1 Guest Passenger Elevators

Rated load (minimum)	1600kg. (3500 lbs.)
Speed (minimum)	Various (1.0mps [320 fpm]), for the hotels higher than 7 floors – 1.6mps
Control	Variable Voltage and/or Variable Voltage – Variable Frequency
Platform	Recommended: 2200 mm x 2000 mm (7ft. 2 in. x 6 ft. 6 in) Minimum: 1100 wide x 1400 deep
Signals	Illuminating Hall Buttons Combination Hall Lantern & Position Indicator at Lobby Level Hall Lanterns, all other floors Car position indicator with arrows
Car and Hall Buttons	Illuminated – Mechanical Displacement Two Car operating panels (Optional)
Door Operator	Electronic Door Control
Entrances / Doors	Recommended: 1200 mm x 2100 mm / Centre Opening (4 ft. 0 in x 6 ft. 11 in.) Minimum: 800 wide x 2100cm high clear opening suitably enlarged where to provide wheelchair access
Safety/ Special Features	Full Door Dual Infrared Sensor / Safety Edge Intercom (3 Way Cabin, Reception and Machine room) Emergency Light Battery backup / emergency landing device (Battery operated) Fireman Switch / Interface with Fire Alarm Music Speaker Camera Digital display Screen

12.02.2 Room Service Elevators

Rated load (minimum)	1600kg. (3500 lbs.)
Speed (minimum)	Various (1.0mps [320 fpm]), for the hotels higher than 7 floors – 1.6mps
Control	Variable Voltage and/or Variable Voltage – Variable Frequency
Platform	Recommended: 2200 mm x 2000 mm (7ft. 2 in. x 6 ft. 6 in) Minimum: 1100 wide x 1800 deep x 2200 high
Signals	Illuminating Hall Buttons Combination Hall Lantern and Position Indicator at Main Service Level Hall Lanterns, all other floors Car position Indicator with arrows
Car and Hall Buttons	Illuminated – Mechanical Displacement One Car Operating Panel
Door Operator	Electronic Door Control
Entrances / Doors	Recommended: 1200 mm x 2100 mm / Centre Opening (4 ft. 0 in x 6 ft. 11 in.) Minimum 1080 wide x 2100 high clear opening
Safety/ Special Features	Full Door Dual Infrared Sensor / Safety Edge Intercom (3 Way Cabin, Reception and Machine room) Emergency Light Battery backup / emergency landing device (Battery operated) Fireman Switch / Interface with Fire Alarm Camera

12.02.3 Freight Elevators

To be proposed by consultant based on project need and volume.

General Specifications

- Variable Voltage,
- Automatic Levelling,
- Automatic Terminal Stops,
- Group Operating System,
- Fixtures,
- Landing Buttons,
- Hall Lanterns,
- Combination Hall Lantern and Position Indicator,
- Car Position Indicator,
- Telephone Cabinet / Intercom,
- Public Address System,
- Finishes (To be coordinated with ID),
- Car Ventilation (An average of 300cfm Ventilation),
- Car Frame, Safety and Governor,
- Counterweights,
- Roller Guides,
- Ropes,
- Compensation,
- Oil Buffers,
- Spring Buffer,
- Keycard system,
- Lift machine room temperature (Cooling: 24°C (40-50% humidity)- Heating: 21°C,
- Emergency lighting.

12.03. Finishes

12.03.1 Guest Lifts

- Guest lift car interiors be carefully designed in line with lift manufacturer's requirements and must be compliant with interior design.
- Floor finishes to be hard wearing type, i.e., ceramic/stone all as specified by the interior designer,

- Wall finishes to be minimum high-pressure plastic laminate finish consistent with guest lift lobby or main lobby décor, with wall mounted chrome handrails on 3 sides at 80cm and mirror wall above all as specified by the interior designer.
- Lift car doors and frame inside and outside to be brushed stainless steel or finished to interior design specification.
- Lighting to be as specified by the interior designer, lift car ceiling with translucent diffuser and LED lighting above is not acceptable. Lift lighting to have automatic switch off when not in use.

12.03.2 Service Lifts

- Floor finishes heavy duty non-slip vinyl / rigidified stainless steel.
- Wall finishes up to a minimum of 150cm to be rigidified stainless steel panels together with rubber protection buffer and above this can be white enameled steel, including the ceiling,
- Corner protection is to be provided.

12.04. Signage

12.04.1 Lobbies

Emergency notice signage provided adjacent lift call points. Signs must be displayed both in English and the local language. All lifts lobbies shall have signage “Do not use lifts in the event of fire” with clear visibility.

12.04.2 Panels

Lift panels need to have signage indicating locations for all outlets and facilities within the hotel, i.e., reception, bar, restaurant, meetings, etc. all as agreed with RHG D&T.

12.04.3 Promotions

Provide LCD screen for promotional signage/displays.

13. LEISURE AREAS

13.01. Heat/Wet Areas

A safety alarm button needs to be provided in all heat rooms linked to the leisure club reception or reception.

All indoor humid areas such as indoor Pools, Jacuzzis, Spa, etc. to have walls and ceiling finishes with anti-fungal coating.

13.02. Pool Areas

Pool to be heated and/or cooled as required depending on climatic conditions to allow year-round use.

Minimum 4 hours water filtration turnover rate for outdoor pools and 6 hours for indoor pools. Ensure suitable maintenance access to pool plantrooms.

14. LANDSCAPING

14.01. Irrigation

Hose bibs with shut off valves for general water supply are required at the building perimeter, main entrance and parking areas. For areas subject to freezing provision is to be made to be able to drain the supply pipes.

Irrigation sprinklers systems are to be agreed with RHG D&T depending on the extent and location of landscaped areas. Where installed, an automatic system using grey water is requested, wherever possible.

14.02. Grey/Rainwater Harvesting

Grey water from guest room showers and wash basins shall be separately piped (away from the toilet foul wastewater) to collect this water centrally ready to be cleaned via recycling system and stored for other usages such as irrigation or supplied to toilets cisterns.

An assessment should be made of the potential for rainwater harvesting and water used for such purposes as irrigation, particularly in water stressed countries.

15. RESPONSIBLE BUSINESS

15.01. Metering & Monitoring Strategy

In order to measure and monitor the usage of all incoming utilities (electricity, water, sewage, gas, district heating/cooling, etc.), appropriate primary Smart meters shall be fitted by the supplier with a standard output facility and connected to the hotel's BMS enabling the hotel to automatically measure the hotel's primary usage. Similarly, separate electronic sub-meters shall be installed to account for usage by individual cost centers in the Hotel: -

Area	Electricity	Cold Water	Hot water	LPG Gas	Heating	Nat Gas	Oil
Guestrooms	X (1 for all)	X (1 for all)	X (1 for all)				
Apartments	X	X	X		x		
Kitchen	X	X	X	X		X	
Kitchen extraction	X						
Laundry	X	X	X				
Leisure Centre/Spa /Swimming Pools	X	X	X				
AHUs (separate metering)	X						
Standby/Emergency Generators	X						X
Gas Boilers						X	
Oil Boilers							X
Total Hot Water hotel			X				
Chillers	X						
Cooling Tower		X					
Total Hot water supply & secondary return temperatures			X				
Water recharge system		X	X				
Chilled Water Flow & Return temperatures from hotel		X					
Irrigation system		X					
Cooling system server room	X						
Tenants' areas	X	X	X		X		

All meters to be connected to the BMS (BACnet, M-bus or Modbus communication protocols) and be accessed remotely. Provide two-way communication between the meter and external networks (both supply & grid) for maintenance and control of the meter. Provide external power supply and backup power supply with built-in charging circuit.

All meters and submeters to be easily accessible and readable.

Individually meter chilled water to each occupancy.

Individually meter heating hot water to each occupancy.

Individually metering in all tenants' areas.

Installation of wireless meters in case cabling is not cost efficient.

15.01.1 Electricity smart meters

Install primary electrical meters according to IEC 62055-41.

Install electrical meters in main switchboards and panels:

Provide digital multifunction meters on main distribution feeders from the main and distribution.

switchboards. Meters provide direct readouts with network connections for networked information accessed from a centralized monitoring location.

Information provided includes:

- Meter ID.
- Voltage (all phases).
- Current (all phases and neutral).
- Wattage (all phases and neutral).
- KW·H consumed.
- Power Factor.
- Suitable IP class depending on the environmental conditions.
- Labelling (reference to the measured system).

15.01.2 Water smart meters

Information provided includes: -

- Meter ID.
- Consumption (m³).
- Suitable IP class depending on the environmental conditions.
- Labelling (reference to the measured system).

15.01.3 Energy smart meters (Heating, cooling etc.)

- Meter ID.
- Consumption (kWh/MWh).
- Measuring the temperature between the supply flow and the return flow.
- Suitable IP class depending on the environmental conditions.
- Labelling (reference to the measured system).

15.01.4 Gas meters (natural gas)

- Consider safety precautions and installation requirements prior to installation.
- Meter ID.
- Consumption (m³).
- Labelling (reference to the measured system).

The following features shall be provided by the BMS in regards with the monitoring of the meters: -

- Monitoring and recording (hourly, daily, weekly, annually).
- Compare consumption for the recorded periods for trend analysis.
- Determine “out of range”/outliers’ values and alert building maintenance operator.
- Record peak energy consumption.

All the pre-mentioned meters shall support remote firmware upgrade feature for meter firmware without loss of any data for a part or complete firmware of meter. Meters shall support the capability to self-register the meters with new firmware.

15.02. Electrical Chargers

Charging station to be conformed with the following European standards/directives: -

- European Energy Performance of Buildings Directive.
- IEC 61851 (Electric Vehicle Conductive Charging System).
- EMC Compliance FCC Part 15 Class A.

The following design parameters to be considered: -

- A minimum of Mode 3 or equivalent.
- Minimum number of ChargePoint’s to be in line with the local regulations.
- Each charging station shall operate independently.
- Have a minimum rated output of 7 kW, measured or calculated at a nominal supply voltage of 230VAC.
- Consider upstream protection and measurement of electricity consumption.
- Operating Humidity Up to 95% non-condensing.
- Operating Temperature -30°C to +50°C.
- Be fitted with a charging equipment status indicator using lights, LEDs or display.
- Electrical safety requirements to be considered as per the latest EN standards.
- Electric vehicle charging equipment shall be supplied by a dedicated branch circuit that supplies no other loads except ventilation equipment intended for use with the EV charging equipment.

- Consider the mounting requirements (pole-mounted, anchored to a concrete base, etc.) and the length of the charging cable in relation to the typical location of EV charging sockets.
- Consider information/indicate signs for parking areas with EV charging stations.

In resorts, placement of several stations for golf cars charging is to be provided as per resort Layout and mapping. Engineering Workshop area to have charging station for engineering electrical maintenance car.

Typical AC Charging Capacity:

The suggested minimal charging capacity per type of hotel and for priority countries is indicated below. Each location is unique. Exact numbers need to be confirmed by a site analysis. Below table indicates the number of AC poles, noting that 1 AC pole can charge 2 EVs simultaneously.

Country	Parking Spots	50	100	150	200	250	300	350	400
Norway	70%	3-4	6-8	8-14	14-18	15-20	20-25	25-35	35-40
Netherlands, Denmark, Sweden, Switzerland	70%	1	2	3-4	3-5	4-6	5-7	7-9	9-12
Austria, France, Germany, Belgium, UK, Russia	70%	1	1	2	2-3	3-4	3-4	4-6	6-7
Portugal, Finland, Estonia, Italy	70%	1	1	1	1-2	1-2	1-3	2-3	2-4
Latvia, Lithuania, Poland	70%	1	1	1	1	1	1	1	1-2

16. ANNEX I - MEP Equipment – Approved Brands

16.01. Electrical

- **Generator:** ELECTRA MOLINS - CATERPILLAR - FINANZAUTO - SDMO / CUMMINS - FG WILSON
- **Power transformer cabinet:** SCHNEIDER - SIEMENS - ABB - TESAR GULF TRANSFORMERS
- **Electrical panels, protections:** SCHNEIDER - ABB - SIEMENS - MERLIN GERIN - EATON - HAGER - GARO – MEGACON - LEGRAND
- **Capacitor bank:** SCHNEIDER - SIEMENS - ABB - CIRCUTOR- LIFASA SPAIN - NOKIAN FINLAND - HITACHI
- **Electric meters, network analyzers:** CIRCUTOR - SCHNEIDER - SIEMENS - ABB - HAGER - MERCURY
- **SAI:** SCHNEIDER - SALICRU - MEGALINE NEWAVE - MILLETEKNIK
- **Transformers (dried, encapsulated and low noise level):** ABB - TRIHAL - SIEMENS - TESAR GULF TRANSFORMERS - SCHNEIDER
- **Electrical mechanisms:** BTICINO - SCHNEIDER – SIMON - SIEMENS.
- **Scenarios ignition rooms:** SIEMENS - BTICINO - LEGRAND - KNX - JUNG - SCHNEIDER - Brands compatible with KNX/Zigbee/Dali communication protocols
- **Lamps and Lamp fittings (LED only):** PHILIPS (Masterline Series) - OSRAM - SYLVANIA - NARVA - GAUSS - FARO - LEDC4 - AROMAS DEL CAMPO - LINEA LIGHT – ROMALUCE - IGUZZINI
- **Motion detectors:** ORBIS DICROMAT - INGENIUM SR – HAGER - SCHNEIDER - ABB
- **Control systems of scenes and lighting level:** TROLL - LUTRON - DALI - SCHNEIDER - WAGO
- **Kitchen Lighting (low energy consumption):** GEWISS - PHILIPS – OSRAM - GAUSS.
- **Light point of the kitchen hood (Anti-deflagrant):** PHILIPS - OSRAM.
- **Electric outlets for kitchens and wet rooms:** GEWISS - BTICINO - SCHNEIDER - ABB - LEGRAND
- **Flame retardant type and halogen free wiring:** PRYSMIAN (Afumex type) - DRAKA CABLES - GENERAL CABLE - DUCAB UAE - RIYADH CABLES - OMAN CABLES - NEXANS - SEVCABEL
- **Supply cables for emergency equipment (Fire):** PRYSMIAN (Afumex Firs) – DRAKA (Cables Protech 1.) – FIRECELL UK – PIRELLI GOLD FRANCE – NEXANS.
- **Metal Trays:** Hot dip galvanized PEMSA REJIBAND - PVC UNEX PC+ABS (halogen free)
- **Protection Pipes:** GEWIS (ICTA series) for flexible hose and (RKHF series) rigid - BARTON UK - SWIFT UK - ACKERMAN UK.
- **Surge Protection:** INGESCO - CIRPROTEC
- **Lightning Rod:** INGESCO - FURSE UK - ERICO USA - WALLIS
- **Telemetry:** SATEL
- **ATS (Automatic Transfer Switch):** ASCO - ZENITH - SCHNEIDER
- **BUSBAR:** EATON - BARDCUT - TRANSLITE
- **VFD:** SCHNEIDER - ABB - TELEMECANIQUE
- **UPS:** EATON - LIBERT - CHLORIDE - Legrand - APC – IPPON
- **CHP:** 2G - GOGENCO - ENERGIE

16.02.Plumbing

- **Pumps:** WILO - GRUNDFOS - EBARA - ITT BELL & GOSSET - KSB - CALPEDA - LOWARA - ARMSTRONG - KSB
- **Dosing and Softening Equipment:** CILLIT - BWT - METITO - VEOLIA - CULLIGAN - GRUNDFOS
- **Automatic self-cleaning Filter:** CILIT (Multipur A)
- **Volumetric electronic softening equipment:** CILLIT (PARAT or BA-PILOT when consumption is very important in dishwashers).
- **Equipment certified to work with water up to 80 °C:** CILLIT HWZ-75
- **Uprights material (Heat fused PPR):** ITALSAN NIRON – POLYMUTAN - WAVIN – NIRON ALU
- **Insulation:** ARMAFLEX - KAIMANFLEX. All insulation joints shall be glued with impact-glue and taped impact by appointment of the same brands.
- **Distribution manifold:** GIACOMINI - ESBE - DAHL - Danfoss - TTM Produkter AB
- **Cross-linked Polyethylene piping:** WIRSBO
- **Water Savers:** RST or similar (Long-Life 24xx models for model RST141x aerators and shower for volumetric reducing flow) - ZURN - PPP - WATTS - WADE
- **Bathroom accessories:** EUROTEC (model as signed in the quality report) - HATTERSLEY - CRANE - NIBCO - PEGGLER - HANSGROHE - GROHE
- **Defogging resistance:** KINNEGRIP (model N2003 55 W) or similar
- **Sensor or timed taps:** GROHEDAL or similar.
- **Soundproof material:** FERROPLAST - TERRAIN - COLD PHON - ABN
- **Manifold soundproofing** (pickup on the ground floor and common areas): FONOBLOK (ARMACELL) - PKB2
- **Grease traps:** CAINOX - KESSEL - COLLINET (PEHD) - ACO
- **Water meters:** MABECONTA - MINOL- ABB - KAMSTRUP - SCHLUBERGER - HERSEY - AQUAMETRO - ITELMA - VALTEC - PULSAR
- **Water heaters:** ARISTON - VIESSMEN - A.O. SMITH - HAASE
- **Water treatment / filtration:** METITO - VEOLIA - CULLIGAN - ASHLAND - ATG UV Technology - ECODAR
- **AIR COMPRESSOR:** ATLAS COPCO - KAESER - COM AIR
- **Mixing Vessels, Accumulator tanks:** ARMATEC
- **Valves:** CIMBERIO - ARMATEC - IMI HYDRONICS AB - DANFOSS - OVENTROP - TECOFI
- **Pipes:** THERMOTECH - UPONOR - REHAU - VIEGA
- **Heating exchangers:** ALFALAVAL - SEDICAL

16.03.Fire safety systems

- **Fire alarm control panel:** NOTIFIER (ID 3000) - SIEMENS (SINTESO FS20 FC2060) - TYCO - SIMPLEX - HONEYWELL GENTS - ADVANCED ELECTRONICS - SECURITON - ESSER - BOSCH - SCHRACK - EDWARDS (Carrier)
- **Fire detection:** SIEMENS CERBERUS - NOTIFIER - TYCO - SIMPLEX - HONEYWELL - APPOLO - HOTCHIKI - SIOX SOLUTIONS - SECURITON - COOPER SAFETY FULLEON - ABB - EDWARDS (Carrier)

- **Fire dampers:** KOOLAIR - SCHAKO - STOC - MADEL - TROX - RUSKIN - SAFID - KBE - BEVENT RASCH - EKOVENT
- **Engines (dampers):** BELIMO (AF series) - KBE - SAFID - ACTION AIR - BEVENT RASCH - HAGAB
- **Recessed fire cupboards:** NOHA (model 3)
- **Fire pump:** GRUNDFOS - WILO - EBARA- AURORA - PEERLESS - PATTERSON - SPP - ITT BELL & GROSSET - Diesel Engine: CATERPILLAR - CLARKS - CUMMINS - FIREDRIVER
- **Emergency lightning:** DAISALUX (HYDRA series) - CEAG - INOTEC - PRAIZZIA - SCHNEIDER - LEGRAND - GAUSS
- **Beacon lights:** DAISALUX (SERPA)
- **Photoluminescent evacuation signs:** JALITE - CEAG - INOTEC - SAFETY SHOP - S10 SUPPLIES - DIRECTA UK
- **Emergency exit light signs:** DAISALUX (VIR series) CEAG - INOTEC - PRAIZZIA - JLB - ECOLUX - GELIA - SCHNEIDER - LEGRAND
- **Approved mortars** (plaster, perlite and Fiberglass CPG): EUROQUIMICA - HILTI - PROMAT
- **Intumescent Collars:** PYROPLEX - ENVIROGRAF
- **Fire expansion collars:** URALITA - WAVIN - ENVIROGRAF
- **Sprinkler Alarm Bells:** VIKING- MARSHAL - NAFFCO - CHUBB
- **Sprinklers:** VIKING (The hidden ones will be MIRAGE) MARSHAL - NAFFCO - CHUBB - TYCO GRINELL
- **Flow indicators:** VIKING MARSHAL - NAFFCO - CHUBB
- **Fire stop material:** 3M - HILTI - ASTROFLAME - ABESCO - NULLIFIER - ENVIROGRAF - SIKA
- **Fire suppression system:** ANSUL - MINIMEX - NAFFCO - NOBLE - NOVEC 1230
- **Foam system:** VIKINGS - SIEMENS - CHEMETRON - FIKE - ANGUS - TYCO - CHEMGUARD - ANSUL
- **Fire Doors:** PROLOC - DALOC - ASSA ABLOY (Besam)

16.04.HVAC installation and DHW production

- **Boilers:** VIESSMANN (Vitocrossal 300) - BUDERUS (Logano) - RENDAMAX (R600) - CLEAVER BROOKS - ECOTHERM - COCHRAN - FULTON - BUDERUS - TURBOPOWER - ARISTON - HOVAL - HAMWORTHY
- **Cooling units:** DAIKIN - TRANE - AERMEC - CARRIER - YORK - MCQUAY - SWEGON
- **Solar panels:** VIESSMANN (VITOSOL 200) - BUDERUS - SOLAHART - BOSCH - EXOTHERM - GIORDANO – JINKO - HEGATECH
- **Accumulators:** LAPESA - VIESSMANN
- **VRV Equipment:** DAIKIN - MITSUBISHI ELECTRIC - LG - SAMSUNG - TOSHIBA - RUSCLIMAT
- **Equipment cooling towers water treatment:** CILIT - VEOLIA
- **Equipment Split:** MITSUBISHI ELECTRIC - DAIKIN - LG - GREE - SAMSUNG - TOSHIBA
- **Air curtains:** THERMOSCREENS - S&P - BERNER - FRICO - DCM - TOSHIBA - TEPLOMASH - TEDDINGTON
- **Pipes:** Polypropylene by thermofusion: ITALSAN NIRON - POLYMUTAN - WAVIN - NIRON - ALU - K&S - SPP - NIPPON - GRINEL - CRANE - BSS - UNOPOR - THERMOTEC - REHAU - VIEGA
- **Insulation:** ARMAFLEX - AF (with steam barrier for cold fluid) - SH (for heating) - KIMMCO - KNAUF - K FLEX - ARMAFLEX - KAI FLEX

- **Pumps:** WILO - GRUNDFOS- ARMSTRONG - KSB - CALPEDA ITT BELL & GROSSET
- **Benches:** MITSA
- **Acoustic Panels BE-PT:** BÁLTICO - STOC
- **Outside plates heat exchanger:** ALFA LAVAL - SEDICAL - SONDEX - SWEP - TRANTER
- **Cathodic protection:** GULDAGER
- **Thermostatic valves:** TA-THERM
- **Regulating valves:** TOUR & ANDERSSON - HATTERSLEY - CRANE - PEGGLER - DANFOS - TECOFI - OVENTOP
- **Towel Radiator:** ROCA - RUNTAL - EUROTEC
- **Damping elements:** MITSAG series
- **Conduits** (according to local normative): CLIMAVER PLUS - URSA AIR ZERO
- **Supply and return grilles:** SCHAKO - AIRFLOW - KOOLAIR - AIR MASTER - TECHNICALCO - BETA
- **Motorized Dampers:** SCHAKO - AIRFLOW - KOOLAIR - SWEGON - BEVENT RASCH - BELIMO
- **Fan-coils** (EUROVENT certified): AIRLAN / AERMEC (FCX Series-P) - GEA - SHAKO (Aquaris Silent SC and SP) - DAIKIN (FWM) - CARRIER - SINKO - TRANE - CARRIER - YORK - ABILITY - CIAT
- **Air handlers:** AIRLAN - AERMEC - GEA - SWEGON TROX - EUROCLIMA - MEKAR - TRANE - CARRIER - YORK - CIAT - REMAK
- **Air Diffusion:** SCHAKO - KOOLAIR - KRANTZ - SWEGON - AIR MASTER - TECHNICALCO - BETA
- **Extraction hoods up to 400 ° C (kitchen ventilation):** S & P - LUIS CAPDEVILLA - SODECA - KOOLAIR
- **Cold rooms:** TAVER - KIDE - SILENSY (for low power) - WILLIAMS - FOSTERS
- **Continuous recording system cold storage:** ELIWELL TELEVIS
- **Rubbish room equipment:** SILENSY type (cubic evaporator with gas R-134a)
- **Meters:** SEDICAL- SCHNEIDER
- **Deaerators:** PNEUMATEX range VENTO
- **Close control unit (ccu server):** LIEBERT - AIREDAL - UNIFLAIR - COMPU AIR
- **VAV:** TUTTLE & BAILEY - TITUS - TROX - SCHAKO - PRICE
- **Air filters:** AAAF - CAMBRIDGE - GLASSFLOSS - CAMFIFAR - AEROFIL - CAMFIL - WESTBURY FILTERS

16.05.BMS and Building Automation Systems

- **BMS and Room Management System** (room controllers, thermostats, actuators,...): JOHNSON CONTROL - KIEBACK&PETER - SIEMENS - HONEYWELL (TREND) -- SCHNEIDER - SAUTER.
- **SMART ROOMS:** PHILLIPS/SIGNIFY, JOHNSON CONTROL - KIEBACK&PETER - SIEMENS - HONEYWELL (TREND) -- SCHNEIDER - SAUTER

16.06.Lifts

- **Lifts:** OTIS - THYSSEN - SCHINDLER - KONE - MITSUBISHI

16.07.Security

- **Cameras:** JVC - PELCO - HONEYWELL - BOSCH - AXIS - AVIGILON
- **Video intercom:** FERMAX (Cityline series) - MILESTONE XPROTECT (RHG approved supplier)

- **Automatic doors:** BESAM - MANUSA - PORTIS - GEZE - BOON EDAM - DORMA
- **Automatic door closers:** GEZE - NORTON - YALE - STANLEY - DORMA - CISA - HAFELE
- **Parking Control:** COSIENSA - PARKING EYE
- **Door's alarm:** ADEMCO - SIEMENS
- **Door lock system:** VINGCARD (ASSA ABLOY) - KABA
- **CCTV recording equipment:** HEITEL (CamDisc 10) - PANASONIC - SENSOR VID - SECURE LOGIC - CUBITEC - VIGILON

16.08. Telecommunications

- **Sound equipment** (Announcement system in lounge and common areas): SONELCO - EIS - BOSE - NEWTEC (Speaker decorative)
- **Room announcement system:** Passive speaker dim.2" EIS - SONELCO - Digital regulator SONELCO (model PCP1221) - EIS (Series 100) - Series BTICINO - SCHNEIDER (as the rest of room)
- **Disabled Toilet Alarms:** ABB - HONEYWELL
- **Sauna control units:** TYLÖ

16.09. Additional brands applicable ONLY for EERUT:

(Eastern Europe, Russia, CIS & Turkey)

16.09.1. Electrical

- **Electric meters, network analyzers:** ENERGOMERA – PULSAR – WIREN BOARD (for internal analysis only), SYSTEME ELECTRIC
- **Electrical Switch Breaker, RCBO, RCCB, MCB, SPD:** EKF (AVERES only), SYSTEME ELECTRIC, IEK (ARMAT only)
- **Generator:** TEKSAN
- **UPS:** SYSTEME ELECTRIC
- **Transformers** (dried, encapsulated and low noise level): SVEL
- **Lamps** (LED only): ECOLIGHT – SVETOVYE TECHNOLOGII – ARLIGHT - GAUSS
- **Kitchens Lighting** (LED only): ECOLIGHT – SVETOVYE TECHNOLOGII – JAZZWAY - ARLIGHT
- **Light point of the kitchen hood** (LED Only): GAUSS – ECOLIGHT – SVETOVYE TECHNOLOGII - JAZZWAY- ARLIGHT
- **Sockets & Switches:** SYSTEME ELECTRIC
- **Busbar:** EKF – IEK
- **Cable tray:** EKF – DKC
- **Thermoindicator:** THERMOELECTRICA
- **Power sockets (for back of house):** EKF, SYSTEME ELECTRIC
- **Controller, Relays:** WIREN BOARD

16.09.2. Plumbing

- **Pumps:** ANTARUS
- **Dosing, Softening Equipment & Water treatment:** ECODAR
- **Water meters:** VZLYET
- **Heating exchangers:** RIDAN – CYCLONE – E8

16.09.3. Fire safety systems

- **Fire alarm control panel:** RUBEZH, SYSTEME ELECTRIC
- **Fire detection:** RUBEZH, SYSTEME ELECTRIC
- **Fire dampers:** ATMOSGRUPP, LUFTMEER
- **Fire suppression system (for kitchen):** KUKHMISTER
- **Emergency announcement:** SONAR by RUBEZH, SYSTEME ELECTRIC
- **Emergency lightning:** GAUSS – ECOLIGHT – SVETOVYE TECHNOLOGII – RUBEZH (evacuation signs)
- **Emergency exit light signs:** RUBEZH – ECOLIGHT – SVETOVYE TECHNOLOGII
- **Fire Doors:** PHOTOTECH
- **Smoke extraction & pressurization fans:** Wheel

16.09.4. HVAC Inst. and DHW Prod.

- **Cooling units:** RUSCLIMAT – WHEEL
- **Solar panels:** HEVEL SOLAR
- **AHU & VRV Equipment:** RUSCLIMAT (incl. SHUFT) – WHEEL – GALVENT
- **Equipment cooling towers water treatment:** ECODAR
- **Pumps:** ANTARUS - CNP

16.09.5. BMS and Building Automation Systems

- **Room FCU thermostats:** HDL, SYSTEME ELECTRIC, WDA (should be agreed additionally depends on a project), BITTEL (should be agreed additionally depending on a project).
- **Room motion sensors:** HDL
- **BMS:** SYSTEME ELECTRIC

16.09.6. Lifts

- **Lifts:** RU KMZ (KMZLIFT) – SJEC

Document Change Control Log

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